Colt Ethernet Services Internal Service Description

Abstract

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Version history and control

Version history and control	Author / Product owners	Approved and reviewed by	Change log: summary of the updates
1.31	Rodolphe Dib	Rodolphe Dib	New note about Dubai in the Demarcation Device paragraph
1.32	Rodolphe Dib	Rodolphe Dib	Version history and control table added
1.33	Rodolphe Dib	Rodolphe Dib	New paragraph about standard handover tests (p.77), Turkey added in comments, 100G Hub spoke limit raised to 200

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1 Ethernet Services Overview

Summary Description of Ethernet Services

Colt's network supports the leading industry-standard optical interfaces and protocols and is based on a redundant and diverse fibre architecture to meet the demands for ever increasing performance and resilience. Ethernet services are available seamlessly across the Metro, National and International areas in 32 countries, 51 metro city networks and over 212 connected cities. The Colt network connects over 900 data centres and 29,000 enterprise buildings globally, including many major business locations.

Colt operates a 49,404 km long distance network, supported by over 9,000km of metro fibre assets. Our ability to provide services end-to-end means that we can offer customers a high quality, cost-effective solution, backed by comprehensive Service Level Agreements (SLAs) and award-winning customer support.

Independent recognition with numerous industry awards and a customer base that includes the world's leading financial institutions, carriers and cloud providers means customers can be confident when they entrust their network to Colt.

2 Benefits

Colt Ethernet services provide many benefits to our customers:

2.1 Business Growth

With over 12.8Tb of 'on-demand' capacity on its optical network, Colt can provide customers with the additional bandwidth to meet business demands efficiently and without lengthy delays.

2.2 Agility

Customers can respond quickly to opportunities and challenges due to Colt's faster lead times. Services up to 40Gbps can be provisioned, meaning customers can get the capacity they need when they need it.

2.3 Strategic Expansion

Directly connecting over 800 data centres and 25,000 business premises, customers can easily expand their network to include new locations as the business demands.

2.4 Cost Efficiency

Colt's high capacity network offering direct DC to DC connections enables Colt to provide attractive commercial offers and support customers in reaching the scalable cost per Mbit they need, regardless of where they are on the capacity requirement curve.

2.5 Business Continuity

Colt delivers its services over a wholly owned and operated infrastructure, thus ensuring the highest levels of Service Availability and backed by Colt's leading Service Level Agreement. As well as targets for delivery, service availability and time-to-repair, the SLA also includes performance guarantees.

A redundant and diverse network architecture enables Colt to provide a comprehensive suite of resilience options, allowing customers to select the resilience solution that best meets their particular chosen balance of cost, performance and risk.

2.6 Performance

Colt's state-of-the-art network provides low latency, low jitter and low frame loss services.

2.7 Market Recognition

Colt's networking products have been independently recognised with numerous awards from prestigious bodies such as the Metro Ethernet Forum and leading industry events such as Capacity Europe. The world's leading financial institutions, carriers, cloud providers and many other enterprises choose Colt for their high capacity backbone network. Customers can be confident when they entrust their network to Colt.

Colt has won many industry awards, including:

- Ethernet Expo Europe, Service Provider of the Year – Best Product Portfolio (2009)
- Global Telecoms Business Innovation Awards – International Optical Service Innovation (2009)
- Global Telecoms Business Innovation Award (2011)
- Global Telecoms Business Innovation Award (2014)
- Global Telecoms Business Innovation Award (2016)
- Global Telecoms Awards - Most innovative cloud service for DCA On-Demand (2017)
- Capacity Europe - Best European Wholesale Offering (2010)
- Global Carrier Awards (2013) Best Pan-European Wholesale Carrier
- Global Carrier Awards (2015) Best Asian Project
- Global Carrier Awards (2016) Best SDN/NFV Deployment
- Global Carrier Awards (2017) Best SDN/NFV Deployment and Best Asian Project
- Leading Lights Winner - Most innovative Ethernet / Optical Service for SDN enabled network (2014)
- Leading Lights Awards - Most Innovative SD-WAN deployment strategy (2017)
- Telco Cloud Forum Awards - Best Telco Cloud SDN/NFV Project for DCNet On Demand (2016)
- Layer 123 Network Transformation Awards - Best SD WAN service & Best new cloud optical solution for DCA On-Demand (2017)

- Layer 123 Network Transformation Awards - Best SD WAN Entreprise service & New Service Innovation (2019)

Metro Ethernet Forum Excellence Awards over a 15-year period

The Colt Ethernet portfolio is certified to MEF CE 2.0

EMEA

2005 Best in Business

2006 Outstanding Innovation

2007 Best in Business

2008 Service Innovation

2009 Service Provider of the Year, Best Service Portfolio

2010 Service Provider of the Year, Best Business Ethernet Service

2011 Service Provider of the Year, Best Business Ethernet Service

2012 Service Provider of the Year, Best Marketing, Best Carrier Ethernet Business Application x 2

2013 Service Provider of the Year, Best Carrier Ethernet Business Application, Best Wholesale Service

2014 Best Service of the Year, Retail Service Provider of the Year, Wholesale Service Provider of the Year, Best Application x 2, Best Network Technology

EMEA & Global Awards

2015 Wholesale Service Provider of the Year (EMEA), Retail Service Provider of the Year (EMEA), Network Innovation (Global), Enterprise Application Awards x 2.

2016 Wholesale Service Provider of the Year (EMEA), Retail Service Provider of the Year (EMEA), Service Innovation (EMEA), Best Marketing (Global) Enterprise Application Awards x 2.

2017 Wholesale Service Provider of the Year, Retail Service Provider of the Year, Network & Service Innovation, Lifecycle Service Orchestration of the Year, Best Marketing, Enterprise Application x 2

2018 Wholesale Service Provider of the Year, Retail Service Provider of the Year, Network & Service Innovation, Enterprise Application x 2

2019 Wholesale Service Provider of the Year, Retail Service Provider of the Year, Network & Service Innovation, Enterprise Application and Lifecycle Service Orchestration (LSO) leadership.

3 Technical Standards Compliance

3.1 MEF Definitions & Standards

The MEF defines the following service types which are matched to the services offered by Colt in the sections below:

Service Type	Port-Based (All to One Bundling)	VLAN-Based (EVC identified by VLAN ID)
E-Line (point-to-point EVC)	Ethernet Private Line (EPL)	Ethernet Virtual Private Line (EVPL)
E-LAN (multipoint-to-multipoint EVC)	Ethernet Private LAN (EP-LAN)	Ethernet Virtual Private LAN (EVP-LAN)

Table 1. MEF Service Definitions

Colt Service	Option Feature	MEF Service Type
Ethernet Line	Port-based	EPL
Ethernet Line	VLAN-based	EVPL
Ethernet Hub & Spoke	Spoke – VLAN-based	EVPL
Ethernet VPN	Any-to-Any / Port-based	EP-LAN
Ethernet VPN	Any-to-Any / VLAN-based	EVP-LAN

Table 2. Colt Ethernet Services and MEF Service Types

Colt Ethernet Services comply with the MEF E-Line, E-Access and E-LAN definition of Carrier Ethernet Services and are MEF CE 2.0 and MEF 3.0 certified.

Colt's engineering function participates fully in technical workshops and in the development of MEF standards. Colt's equipment vendors are also MEF certified.

4 Colt Network Coverage

Colt's secure and reliable network provides unrivalled reach across 32 countries with fibre-based metropolitan area networks (MANs) in 51 major cities. More than 31,000 buildings are directly connected using Colt fibre to this network, and over 1,000 data centres.

4.1 European coverage

Colt's long-distance pan-European transmission network spans over 49,000km of fibre-optic infrastructure, in conjunction with over 9,000Km of metro fibre. Ethernet services are provided within 14 countries: Austria, Belgium, Denmark, France, Germany, Italy, Luxembourg, Netherlands, Portugal, Republic of Ireland, Spain, Sweden, Switzerland and the United Kingdom.

Colt also has PoPs in the Czech Republic, Finland, Slovakia and Norway, connecting key data centres in these locations with the rest of its European footprint.

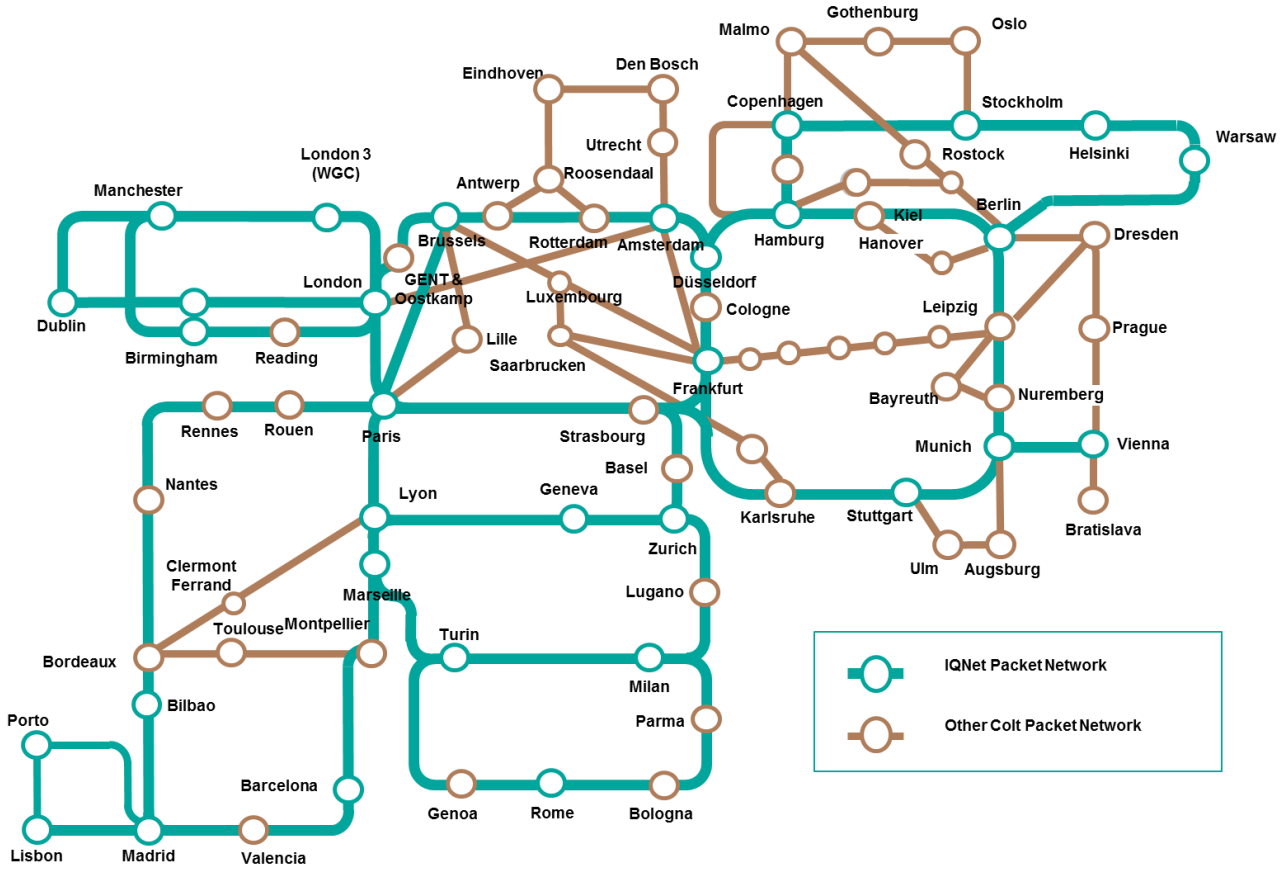


Figure 1. European Long Distance Network

In Europe, Colt has today over 202 city PoPs in 22 countries. The IQNet platform is composed by 170+ PoPs based directly in data centres (Key Data centres), from which services can be directly provided.

4.2 Asia Pacific coverage

Colt operates Packet Network in Japan, Hong Kong and Singapore including access PoPs in many key data centres:

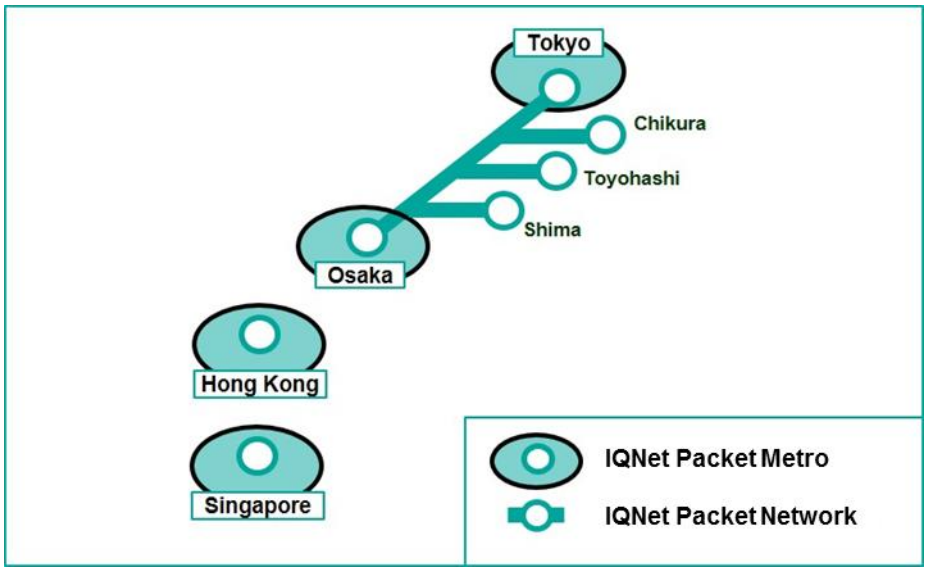


Figure 2. Packet Networks in Asia (April 2017)

The Asia Intercity Packet Network is also based on the Cisco platform. Colt assets is composed by 6 Metropolitan Area Networks (4 in Japan -Tokyo, Yokohama, Osaka and Nagoya - Singapore and Hong-Kong) and 17 city POPs, connecting more than 5000 buildings.

Japan :

- Tokyo (Greater Tokyo) and Osaka metro network
- Tokyo-Osaka national network
- 70+ DCs and 2,800+ retail buildings

Singapore :

- Singapore metro network
- 10+ DCs and 1,500+ retail buildings(90%)
- Implementing own optical fibre network within CBD area

Hong-Kong :

- Hong Kong metro network
- 12+ DCs and 2,500+ retail buildings(80%)

4.3 North-America coverage

The USA is already a part of the Global Reach program with 9 existing DCs in the USA : Los Angeles, Aurora, Chicago, Boston, New-York (x4), Newark supporting the MMSP platform. With IQNet, Colt is still extending its network to enhance its footprint in the USA and also to propose a new country in North-America with the Canada.

The IQnet platform will be installed in 13 data centres (9 new data centres and 3 existing data centres) which will raise the number to 18 connected data centres in North-America where Ethernet Services can be delivered on its own infrastructure and on partners infrastructures.

Region	State	City	Street, Nb	Postal Code	Simple DC Name	Site Owner
US	WA	Seattle	2001 6th Ave	98121	**Westin Building**	Westin Building
Canada	ON	Toronto	151 Front St W	M5J 2Z1	**151 Front (TR1/TR2)**	Equinix
US	MA	Boston	1 Summer St	02110	**One Summer**	Markley Group
US	AZ	Phoenix	615 N. 48th St	85008	**IO.Phoenix One**	IO Data Centers
US	TX	Dallas	1950 N Stemmons Fwy	75207	**InfoMart (DA6)**	Equinix
US	FL	Miami	50 NE 9th St	33132	**NAP of the Americas (MI1)**	Equinix
US	IL	Chicago	350 East Cermak Rd	60616	**Equinix Chicago (CH1/CH2/CH4)**	Equinix
US	GA	Atlanta	56 Marietta St NW	30303	**56 Marietta**	TelX/DRT
US	VA	Ashburn	21711 Filigree Court (campus)	20147	**Equinix Ashburn (DC1/2/3/4/5/6/10/11/12)**	Equinix
US	CA	San Jose	11 Great Oaks Blvd (campus)	95119	**Equinix San Jose (SV1/SV5/SV10)**	Equinix
US	NY	New York	111 8th Ave	10011	**111 8th Avenue**	TelX/DRT
US	NY	New York	60 Hudson St	10013	**60 Hudson**	TelX/DRT
US	CA	Los Angeles	624 S Grand Ave	90017	**One Wilshire**	CoreSite

Table 3. IQNet Data Centres in North-America

The network on the below diagram is planned to go live in April 2018 but is ready for sell as of January 2018. The available services will be Ethernet Line and Ethernet Hub & Spoke.

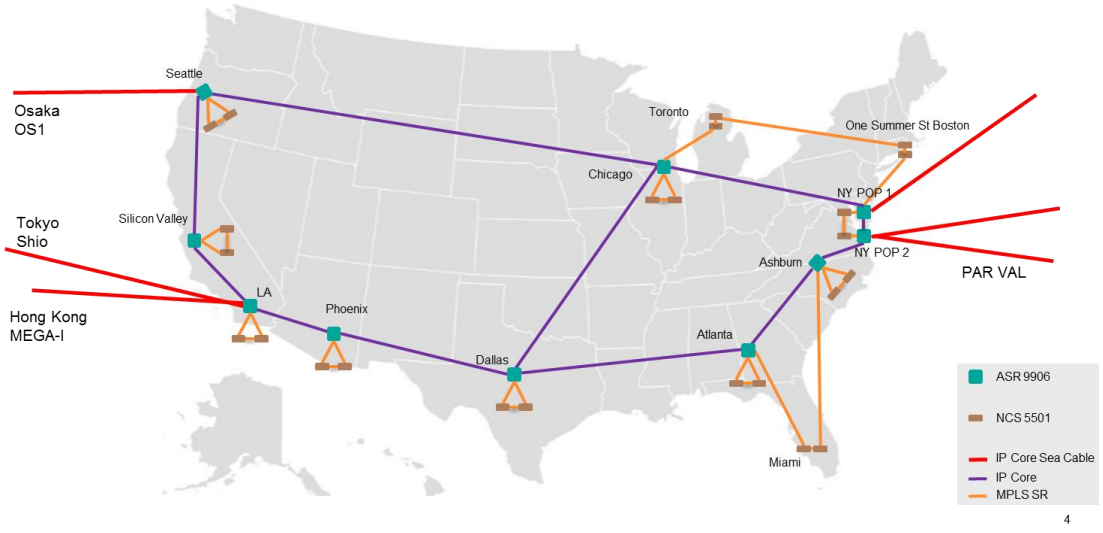


Figure 3. Phase 1 Build (April 2018) 40G packet

Only the connected data centres will compose the on-net part of the network. The retail buildings will be connected using third party partners.

All the 100 Gbps US internal circuits will be hired to third party network operators which implies 10 Gbps bandwidth limitation for national Ethernet Line services.

These 100 Gbps circuits will be shared between Ethernet and Optical services with the following distribution : 60 Gbps for the Optical services and 40 Gbps for the Ethernet services.

4.4 International network

4.4.1 Existing infrastructure

Colt's network already connects three continents: Europe, America and Asia. The connectivity is provided by hiring 10 Gbps capacity links to third party network operator in sub-sea cables.

The current infrastructure supports the MMSP platform.

The diagram below shows the links between Europe and North-America.

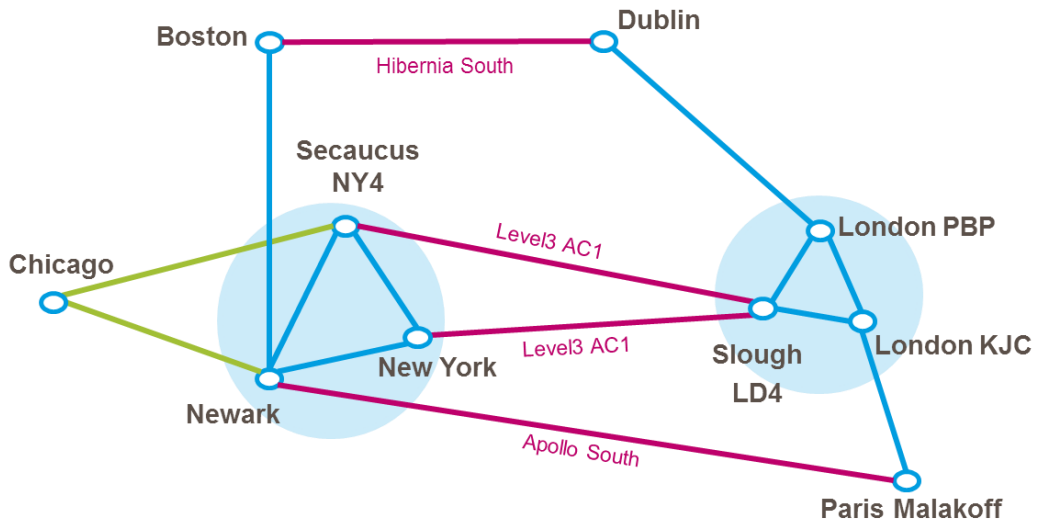


Figure 4. Europe – USA 10 G sub-sea cables (M-MSP)

The diagram below shows the links between Asia and Europe and Asia and North-America.

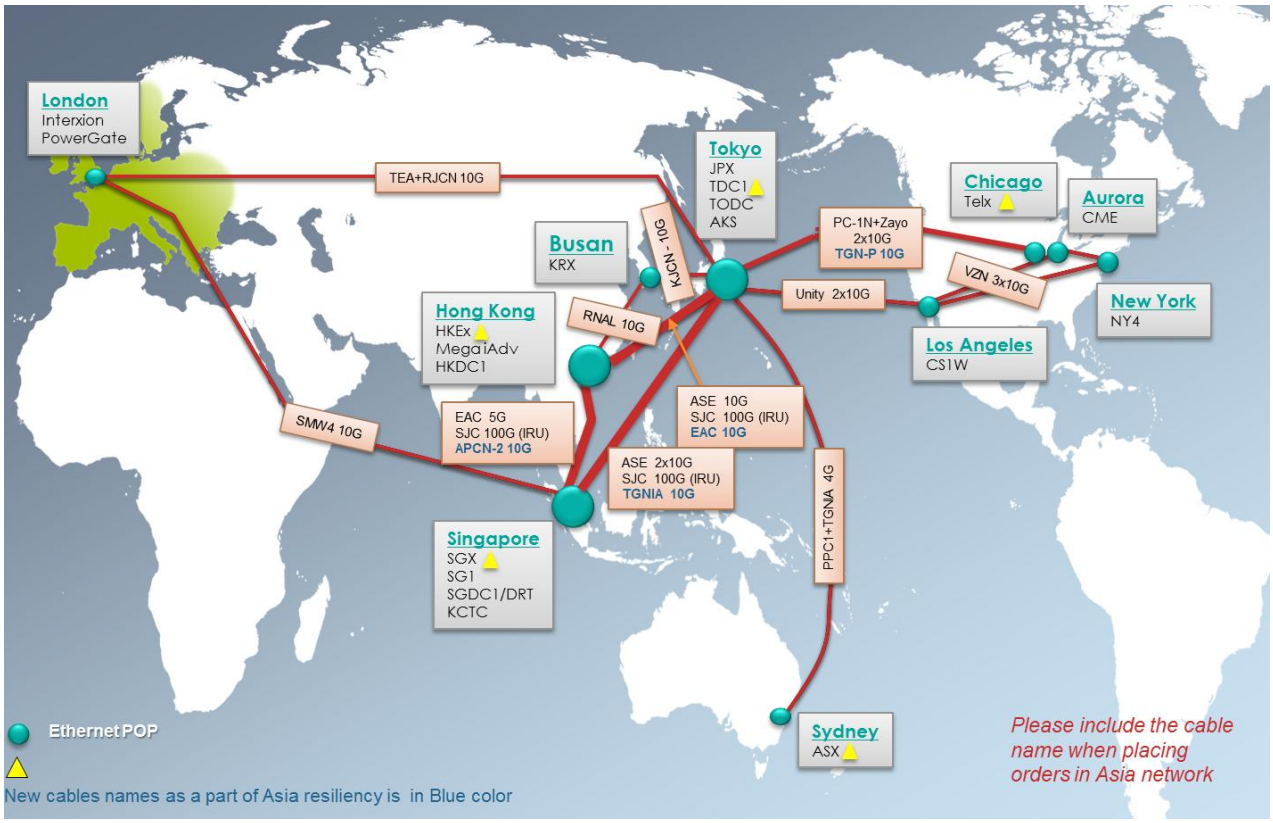


Figure 5. Europe – Asia - USA 10 G sub-sea cables (M-MSP)

The current infrastructure is the legacy of the assets owned by Colt and KVH before the two companies' merger. The target infrastructure will provide a more seamless network as the Colt network, able to provide high bandwidth services.

4.4.2 Target infrastructure

Colt is improving the capabilities of its sub-sea cables by adding new international 100G submarine cables to seamlessly interconnect the three continents. With multiple cable systems and different landing points, Colt will be providing diverse network paths to offer even more resilient solutions to its customers. (The cable system is not selectable for Ethernet Services).

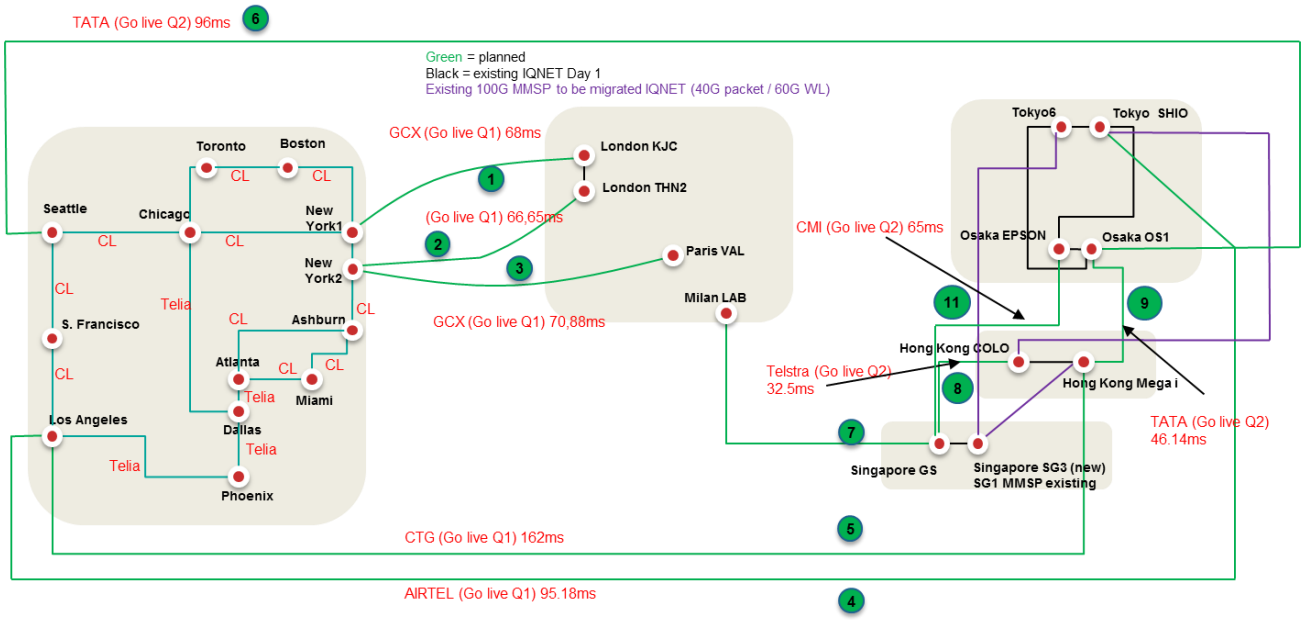


Figure 6. Europe – USA – Asia 100 G sub-sea cables

Aria	Link	City	City	Supplier	Cable System	est. RTD [ms]
Intra-US	Chicago - New York	Chicago	New York	CenturyLink	-	19.1
Intra-US	New York - Ashburn	New York	Ashburn	CenturyLink	-	5.6
Intra-US	Atlanta - Ashburn	Atlanta	Ashburn	CenturyLink	-	11.2
Intra-US	Dallas - Atlanta	Dallas	Atlanta	Telia	-	20.4
Intra-US	Dallas - Chicago	Dallas	Chicago	CenturyLink	-	25.2
Intra-US	Phoenix - Dallas	Phoenix	Dallas	Telia	-	22.8
Intra-US	Los Angeles - Phoenix	Los Angeles	Phoenix	Telia	-	9.5
Intra-US	Los Angeles - San Jose	Los Angeles	San Jose	CenturyLink	-	7.2
Intra-US	San Jose - Seattle	San Jose	Seattle	CenturyLink	-	17.3
Intra-US	Chicago - Seattle	Chicago	Seattle	CenturyLink	-	47.1
Intra-US	New York - New York	New York	New York		DF Packet	0.3
Intra-US	New York - Boston	New York	Boston	CenturyLink	n/a	5.0
Intra-US	Boston - Toronto	Boston	Toronto	CenturyLink	n/a	14.2
Intra-US	Chicago - Toronto	Chicago	Toronto	CenturyLink	n/a	13.6
Intra-US	Miami - Ashburn	Miami	Ashburn	Telia	n/a	20.8
Intra-US	Atlanta - Miami	Atlanta	Miami	Telia	n/a	15.5
Transatlantic	London - New York	London	New York	AquaComms	AEConnect	??
Transatlantic	London - New York	London	New York	GCX	FA-1 North	66.7
Transatlantic	Paris - New York	Paris	New York	GCX	FA-1 South	70.9
Transpacific	Tokyo - Los Angeles	Tokyo	Los Angeles	Airtel	Unity	95.2
transpacific	Osaka - Seattle	Osaka	Seattle	TATA	TGN-P North	96.0
transpacific	Hong Kong - Los Angeles	Hong Kong	Los Angeles	CTG	AAG	162.0
Europe-Asia	Milan - Singapore	Milan	Singapore	Retelit	AAE-1	??
Intra-Asia	Osaka - Singapore	Osaka	Singapore	CMI	APG	65.0
Intra-Asia	Osaka - Hong Kong	Osaka	Hong Kong	TATA	TGN-IA	46.2
Intra-Asia	Hong Kong - Singapore	Hong Kong	Singapore	Telstra	EAC 2A-1	32.5

Figure 7. Europe – USA – Asia 100 G sub-sea cables round trip delay

5 Ethernet Platforms

5.1 General

General description of the Ethernet platforms including reference to the older platform M-MSP, IQNet being the network whom the deployment is ongoing.

5.2 Modular MSP

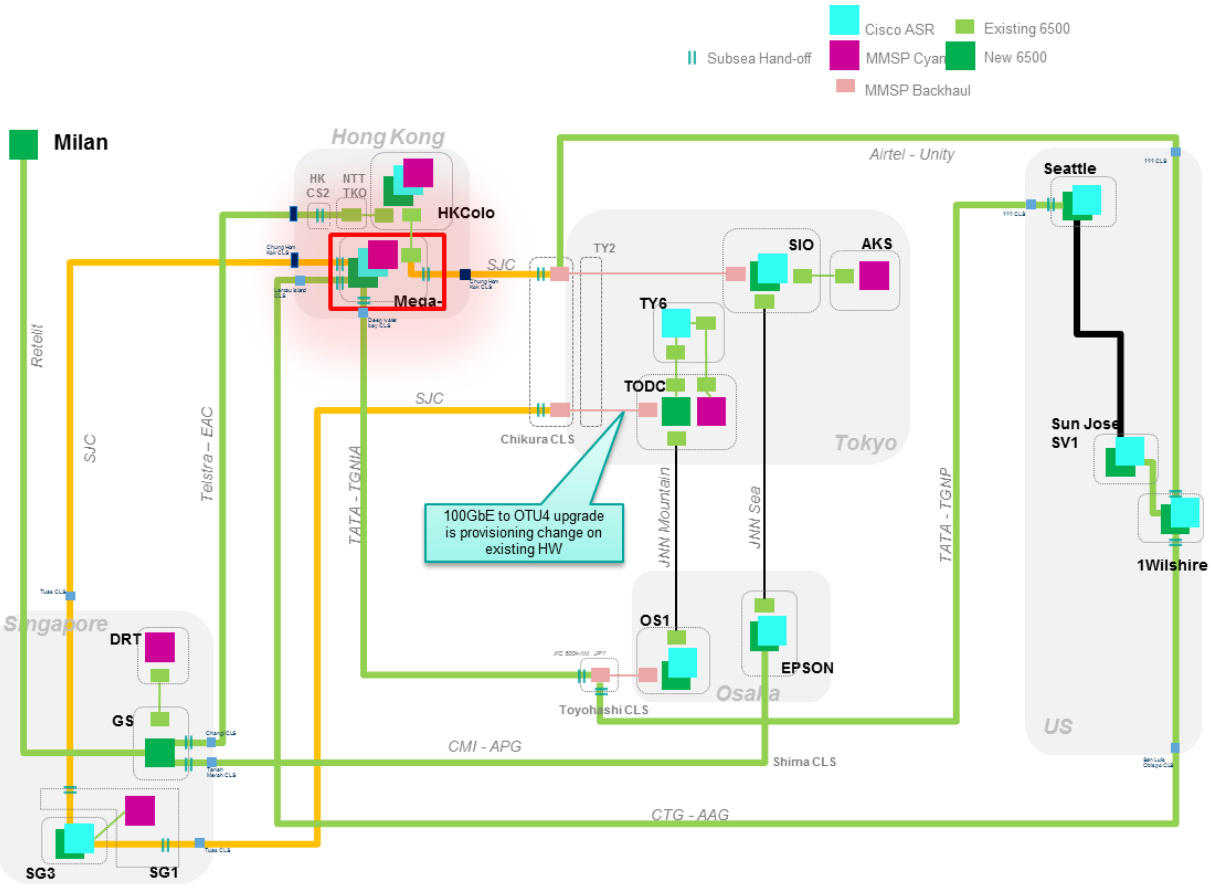
The Multi Service Platform (MSP) is an Ethernet based platform that supports Ethernet Line Point to Point services up to 10Gbps (6 Gbps in Japan), Ethernet Hub and Spoke services up to 10Gbps (6 Gbps in Japan) and Ethernet VPN services up to 10Gbps (6Gbps in Japan). Because it is based on native Ethernet technology, MSP offers lower latency and high levels of flexibility. MSP uses a resilient access ring architecture which is dual homed for added equipment redundancy. MSP is not available in Hong-Kong.

5.3 IQNet - Packet Network

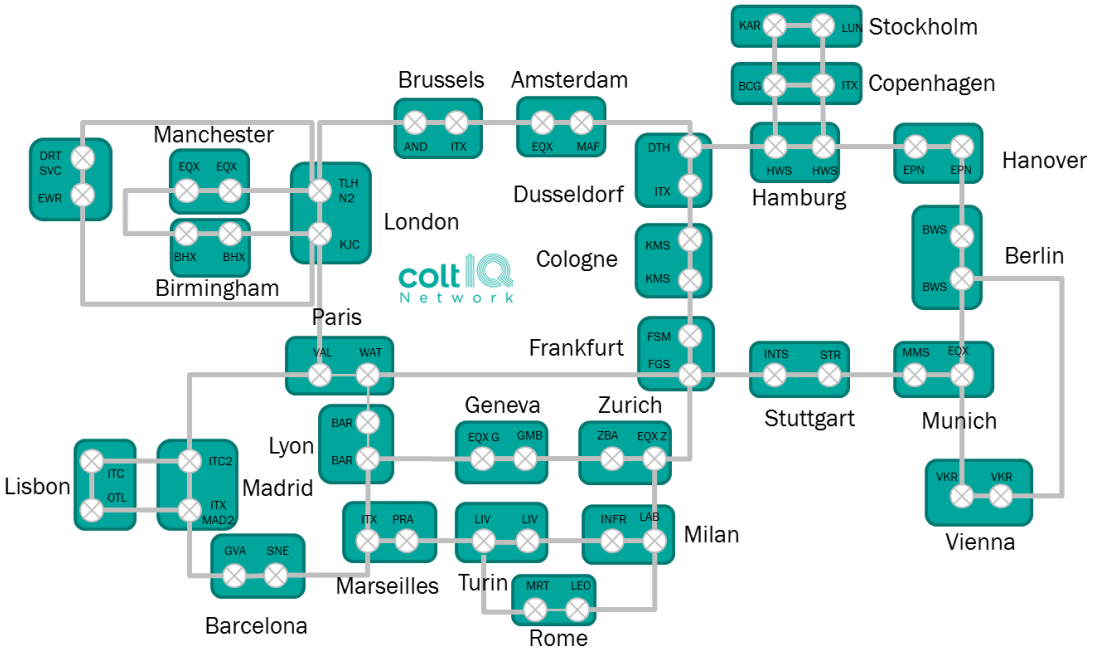
The Colt IQNet network is Colt's advanced packet based network infrastructure supporting Ethernet Line services up to 100Gbps in certain locations, Ethernet Hub & Spoke services up to 10Gbps and Ethernet VPN services up to 40Gbps. Available over Colt's global footprint, Ethernet services are mapped across the IQNet network via fixed connections delivering repeatable performance without variations in routing. Each connection has an optimised protection path in the core network enabling fast protection switching. The platform also supports automated configuration and service enablement ensuring rapid, error-free provisioning.

5.3.1 Network Architecture

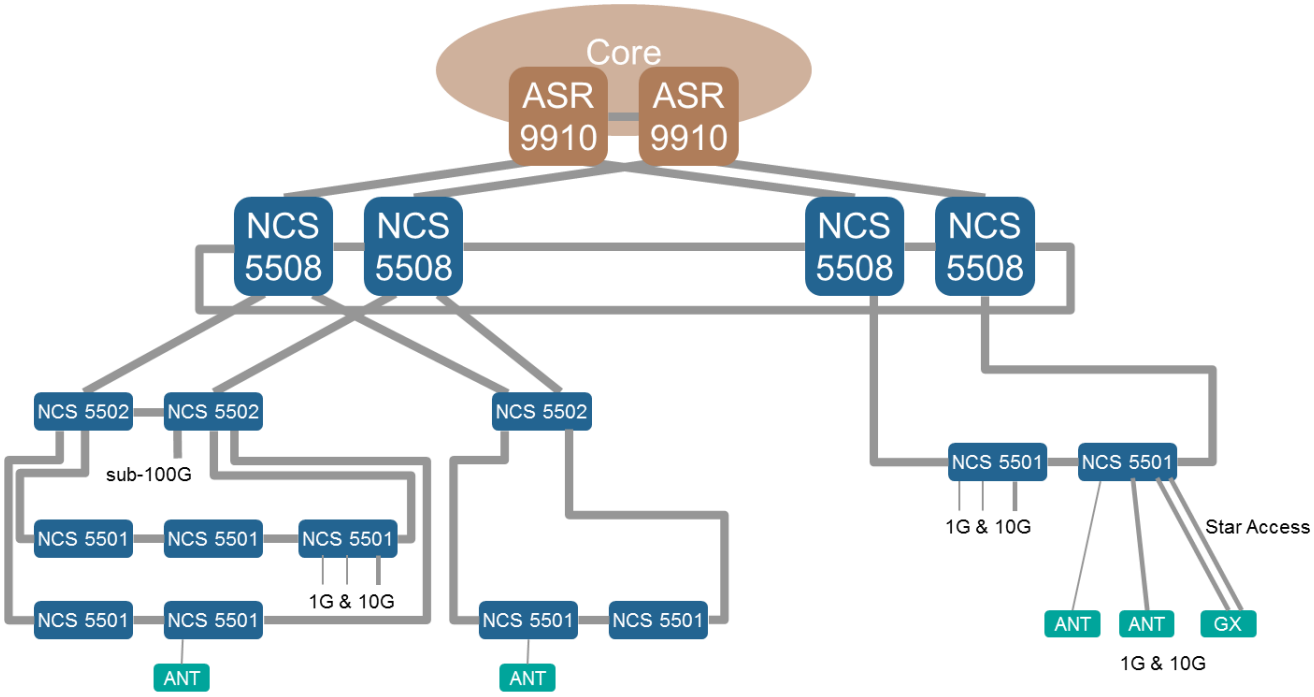
5.3.1.1 IQNet : International backbone



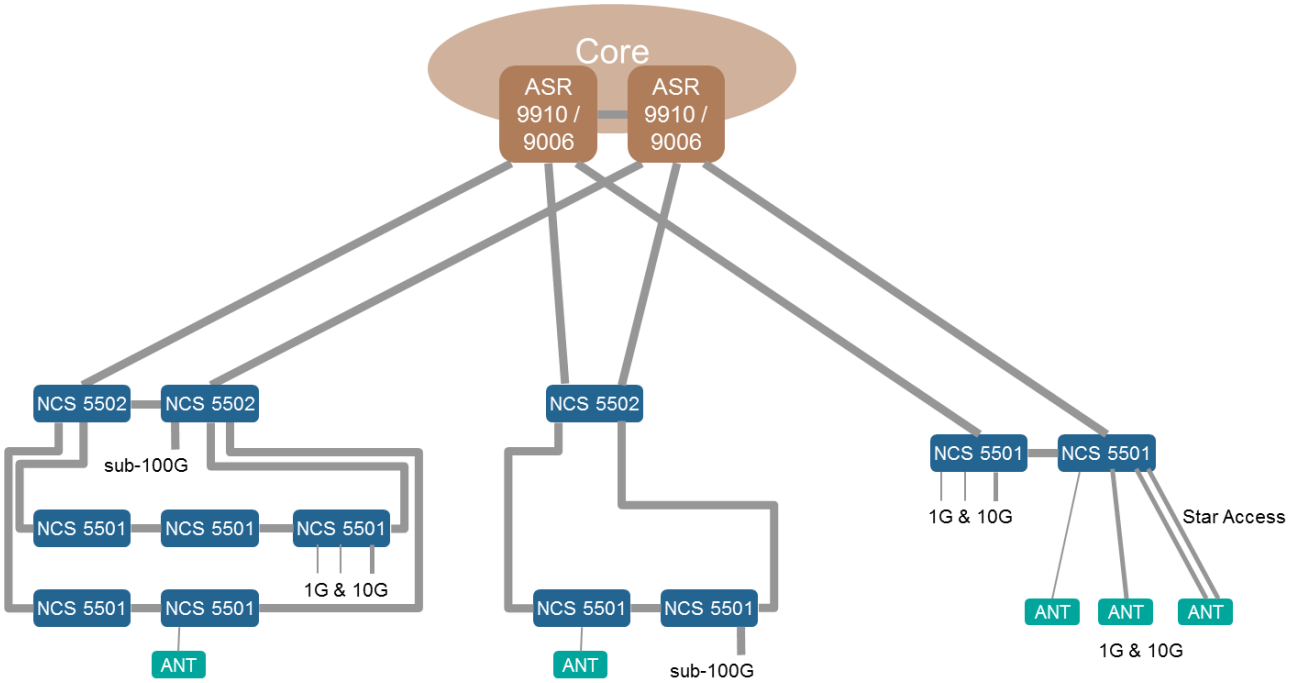
5.3.1.2 IQNet : European backbone



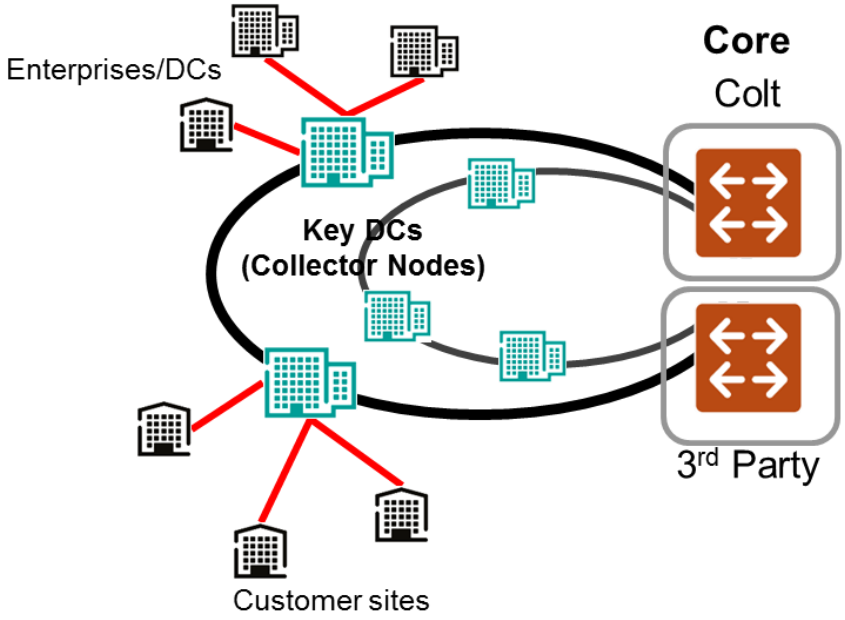
5.3.1.3 IQNet Tier-1 City : Metro backbone



5.3.1.4 IQNet Tier-2/3 City : Metro backbone



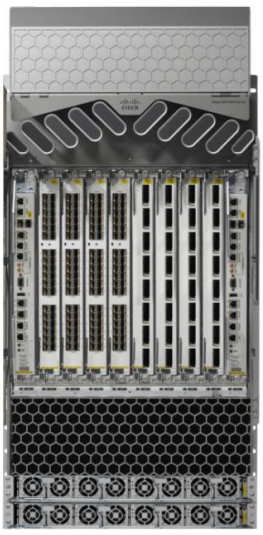
5.3.1.5 IQNet : Access architecture



- Collector Nodes (Key DCs) give deeper Metro coverage.
- Pre-provisioned equipment/capacity guarantees faster lead times in the Key data centres.
- Unprotected services from Collector Nodes to customer premises become default (typically < 3km).

5.3.2 Cisco backbone devices

5.3.2.1 Cisco Tier 1&2 P/PE ASR9910 router



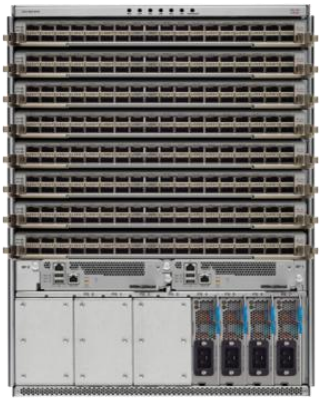
- 21U / 8 slots / 24Tbps ready / 10.7 kW max
- Provides CORE P router & L3 PE function day 1
- Single P/PE pair per city (other Tier 1 metro nodes will use 5508 P routers)
- Only connects NCS5508 / NCS5502 & 5501 / NCS5501 100G edge rings
- 8x 100G PAYG line card default (4 ports paid day1)
- 8k L3 VPN supported (5-6 years Tier 1 volume)

5.3.2.2 Cisco Tier 3 P/PE ASR9906 router



- 10U / 4 slots / 3.2Tbps ready / 5.5 kW max
- Provides CORE P router & L3 PE function day 1
- Single P/PE pair per Tier 3 / POP city
- Only connects NCS5508 / NCS5502 & 5501 / NCS5501 100G edge rings
- 8x 100G PAYG line card default (4 ports paid day1)
- 8k L3 VPN supported (no scaling limit expected)

5.3.2.3 Cisco Tier 1 Metro P router aggregation router NCS5508



- Fixed 13U / 28 Tbps / 7kW max / 8 slots.
- Backbone multiple 100G possible + ready for future dense port 400G LC.
- Scale variant ordered for the Colt build (roadmap L3 PE uplift potential), day 1 P or L2 PE.
- TCAM on board per LC with 2M IPv4 / 1M IPv6.
- Aggregates all Tier 1 5502 & 5501 access rings, switching traffic between metro sections (avoids metro traffic to pass P/PE expensive ASR9K router).
- 3-9 µs local switching max delay.
- LON will have 4 pairs giving us a max of 1536x 100G ports today.
- Day 1 LC 24x 100G using QSFP28 optics (max 10km), using optical Ciena or cube amplifier > 10km (40G break out option supported)

5.3.2.4 Cisco Metro and DC P router aggregation router NCS5502



- Fixed 2U / 4.8 Tbps / 48x 100G ports / 2kW
- Backbone multiple 100G possible
- Scale variant ordered for the Colt build (roadmap L3 PE uplift potential), day 1 P or L2 PE.
- TCAM on board 2M IPv4 / 1M IPv6

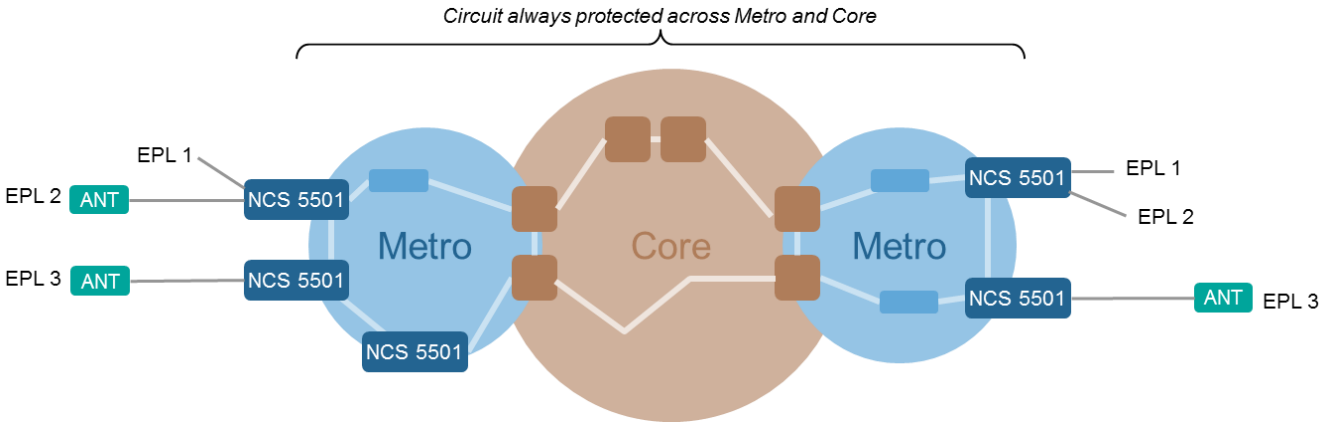
5.3.2.5 Cisco Edge Access NCS5501 router



- Fixed 1U / 1.08 Tbps / 6x 100G & 48 1/10G SFP(+) / 400W max
- Backbone multiple 100G possible
- Tier 1 DC 100G rings (from NCS5502 AG) with multiple 5501 to provide massive port density
- Tier 2 DC stand alone 100G rings with multiple 5501 (connected on 5508 Tier 1 or ASR9K T2-3)
- UNI 1-100 Gbps drop capacity

5.3.3 IQNet drops (not valid anymore – just kept as informational)

5.3.3.1 Drop 1

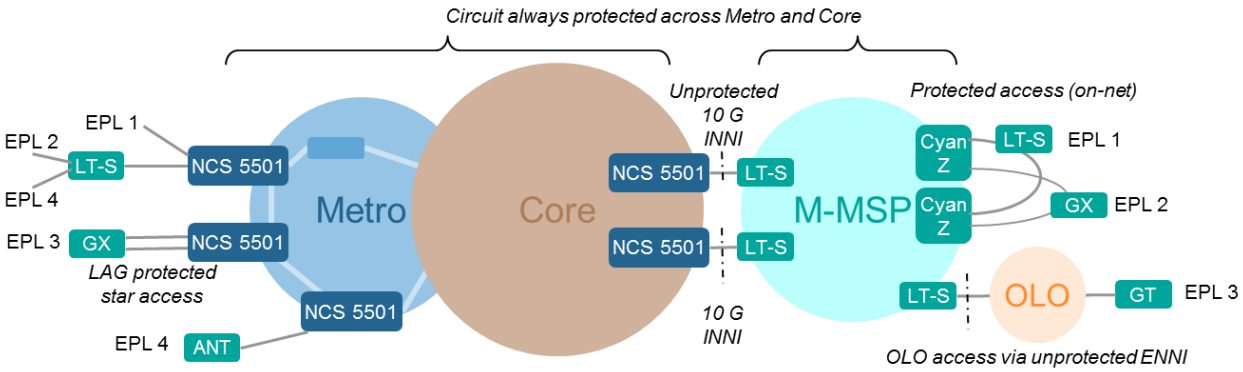


Drop 1 supported Ethernet Line services with both endpoints on IQ Network :

- Direct customer termination on Cisco NCS 5501.
- Star access with termination on ANT 1G.

Note: "Protected/Unprotected" service option on IQ Network always refers to the **access tail** since the metro and core network is always protected (for all services).

5.3.3.2 Drop 2



Drop 2 added support for VLAN based Ethernet Line (EVPL), protected access (LAG) and INNI interconnect with M-MSP :

- All endpoint combinations supported (diagram only shows 3 examples).
- Multiple services per Accedian GX and LT-S.
- Termination options added:
    - M-MSP on-net Accedian GT, GX and LT-S,
    - M-MSP off-net Accedian GT.
- The INNI between IQ Network and M-MSP is unprotected as usual (local link).

EVPL Services

- Only single VLAN ID per circuit supported (NCS 5501 restriction).
- Only one circuit per Accedian ANT.

Unprotected access on M-MSP

- Even if customer orders unprotected access it will be delivered as protected if the site is connected via an access ring.
- M-MSP can also support unprotected star access from Accedian LT-S, but not much used (was developed for fibre ULL).

INNI

- The INNI between IQ Network and M-MSP is always unprotected as all INNIs today.
- Will be used for services with protected and unprotected access.

6 Network Access Types

Colt has a comprehensive range of options for connecting the customer's sites to Colt's services. Whichever access method type is used, the solution will be created and managed by Colt under a single end-to-end Service Level Agreement.

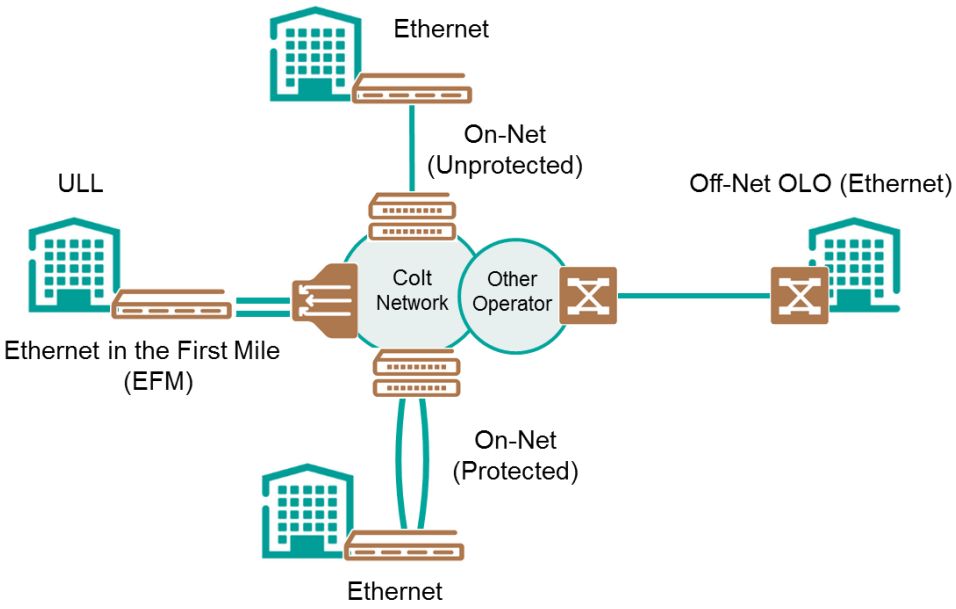


Figure 8. Network access technologies

Customer sites can be directly connected to this network using the following access types:

6.1 Colt Fibre

Offices located in buildings that are directly connected to the Colt network will be connected via Colt's own fibre-optic infrastructure. Colt's Metropolitan fibre networks deliver high capacity, high speed bandwidth services to Europe's major business centres, which in turn are interconnected via the Colt long distance network (LDN) delivering high quality reliable services end-to-end.

Colt fibre is available in Europe and selected buildings in Singapore.

6.2 Unbundled and Regulated Fibre

In a limited number of locations within Europe (Switzerland) and Asia (Japan, Hong-Kong and Singapore) the provision of fibre optic infrastructure in the metropolitan area has been subject to regulation by governmental and regulatory bodies. Colt also uses this fibre unbundled local loop (ULL) to provide network services to customer sites, in conjunction with its own network infrastructure and customer site terminating equipment.

6.3 Ethernet in the First Mile (EFM)

In order to complement the coverage of its fibre network and reach additional buildings, Colt has invested in unbundled local loop (ULL) - copper pair connections from the incumbent operator, available in 6 countries (Belgium, France, Germany, Italy, Spain, Switzerland). Colt uses ULL to provide Ethernet in the First Mile (EFM) connections to customer sites. EFM is a cost-effective access technology based on Ethernet protocol that allows high bandwidth connections over (PTT) copper lines. Bandwidths are available up to speeds of 40Mbps dependent on location and the distance of the building from the CO.

This type of access is only available in European Colt countries.

Note: This Access Type is no longer available for new services as of end 2020

6.4 Partner access types

For sites not directly connected to the Colt network, Colt Ethernet services can be provided using a mixture of Colt and third-party other licensed operator (OLO) infrastructure and services. Colt interconnects with and manages more than 120 partners in order to connect sites that are outside the reach of the Colt network in Europe, Asia or in other global locations.

Colt will order a circuit or circuits from the OLO on the customer's behalf, and customers will still be dealing directly with Colt at all times. Colt also takes responsibility for testing the interface between the Colt network and the OLO's circuit to ensure that the customer's overall end-to-end Ethernet service operates seamlessly.

Please, note that Demarcation devices are installed by default for Enterprise customers while it's an option for Wholesale customers. However, demarcation devices are always installed for any customers to connect off-net Ethernet VPN sites.

Customer sites can be connected through partner networks using the following access types:

6.4.1 Leased lines

Leased lines are circuits or a combination of circuits designated to be at the exclusive permanent disposal of a given customer. Colt can use technically compliant third-party optical services that are themselves provided on WDM platforms or 'clear channel' layer 1 optical Ethernet technologies to extend its own Ethernet services to off-net locations.

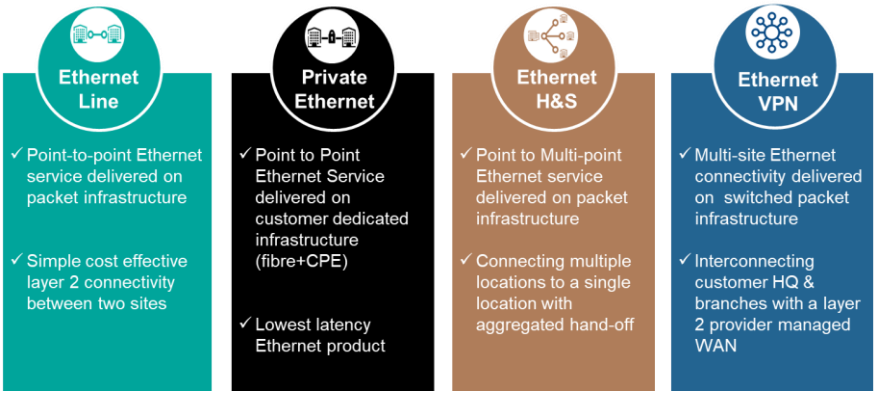
In the specific case of 1Gbps,10G LAN PHY and 100G LR4 services that use Ethernet interfaces, optical Ethernet services of the same presentation type can also be used to extend Colt Ethernet services to off-net sites. The use of such services is subject to Colt's own technical validation process and inclusion in the list of services approved for use with Colt Ethernet.

Colt can also use Ethernet demarcation devices to provide sophisticated standards-based Operations Administration and Maintenance (OAM) end-to-end on services that have a 1Gbps or 10Gbps Ethernet presentation.

7 Service Description

7.1 Service Overview

The Colt Ethernet portfolio of services enables customers to interconnect their Ethernet Local Area Networks (LANs). Ethernet is the de facto industry standard protocol within the LAN, and Colt Ethernet enables customers to extend Ethernet into metropolitan and wide area networks (MANs and WANs). Colt supports multiple Ethernet service options as shown below:



The customer can connect LANs within the same city (Ethernet Line Metro), within the same country (Ethernet Line National) or between countries (Ethernet Line International). The Colt Ethernet Hub and Spoke service, enables the aggregation of Ethernet Line Metro, National and International services onto a single Ethernet interface at a single (Hub) site. Ethernet VPN provides switched access between multiple sites also on a metro, national or international basis. This Any-to-Any option allows all sites of a multi-site configuration to communicate as a virtual private network.

Colt Ethernet Services are transparent Layer 2 services offering:

- Point-to-point, Hub & Spoke and Any to Any topology options
- Speeds from 2Mbps throughput to 100Gbps
- A dedicated service to a single customer
- Metro, National and International coverage
- Unprotected, Protected Resilience Options
- Off-Net access using a range of partner access types

The tables below give an overview of the Ethernet services and features available per region:

Service	Europe	Turkey	Middle-East (Dubai)	Asia	North America
Ethernet Line (Metro, national, international)	√	√	√ (check with Regulatory team for metro and national services)	√	√
Ethernet Hub & Spoke (Metro, national, international)	√	√	√ (check with Regulatory team for metro and national services)	√	√
Private Ethernet (Metro)	√	N/A	N/A	√	N/A
Ethernet VPN (Metro, National, International)	√	√	√ (check with Regulatory team for metro and national services)	√	√

Table 4. Service availability

Features / options	Europe	Turkey	Middle-East (Dubai)	Asia	North America
Resilience (protected)	√	√	√	√	√
Resilience (Gold/Platinum) *(removed from the portfolio)*	*N/A*	*N/A*	*N/A*	*N/A*	*N/A*
Diversity	√	√	√	Bespoke	√
Encryption (Ethernet Line)	√	√	√ (offnet is not allowed)	√	√
Class Of Service [1]	√	√	√	√	√
Link Aggregation	√	√	√	√	√

Link Loss Forwarding	√	√	√	√	√
Proactive Monitoring	√	√	√	√	√
Performance Reporting	√	√	√	√	√
Colt Online	√	√	√	√	√
Demarcation device (OLO)	√[2,3,4]	√[2,3,4]	√[3,4]	√[2,3,4,5]	√[2,3,4]
Fast Track	√	√	√	√ (Hong-Kong only)	N/A
Upgrade Interface	√ (one time charge)	√ (one time charge)	√	√ (MRC)	√ (one time charge)
Dual Power Source	√	√	√	N/A	√
Dual Building Entry	√	√	√	N/A	√

Table 5. Features/Options availability

[1] only available for Ethernet VPN on IQNet

[2] available by default in North America, Asia and in European Colt countries for ENT customers

[3] mandatory for Ethernet VPN on IQNet

[4] Bespoke process required to install a demarcation device for offnet sites in GR countries and Dubai – See the “Demarcation Device” paragraph further down for more details

[5] only supported in Seoul, Busan and Incheon in South Korea

7.2 Ethernet Line

Ethernet Line provides connectivity to customers between two sites. Ethernet Line is compliant with the MEF E-Line definition of an Ethernet Private Line (EPL) service and is a point-to-point Ethernet Virtual Connection (EVC).

Colt Ethernet Line Services are provided on a physical port (EPL) based on a simple service to port mapping, or as one or more VLANs on an Ethernet interface (EVPL).

Port – based services are transparent to customer VLANs or Ethernet frames with headers containing multiple VLAN tags. Either untagged or tagged frames may be transmitted over such links. EVPL services accept VLAN-tagged frames, tagged with the identifying VLAN for each EVPL.

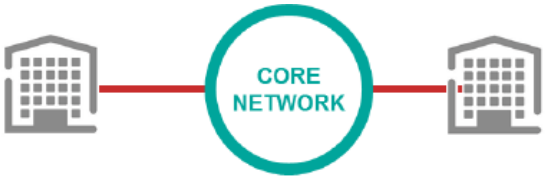


Figure 9. Network Ethernet Line

7.2.1 Ethernet Line Specifications

Item	Specifications
Area	Metro*, National* and International
Service topology	Point-to-point
Service Type	Guaranteed
Service mapping	Map UNI ports to circuits
Interface	100BASE-T 1000BASE-SX/LX/T 10GBASE-SR/LR/ER 100GBASE-LR4
Bandwidth	2M, 4M, 6M, 8M, 10M, 20M, 30M, 40M, 50M, 60M, 100M, 200M, 300M, 400M, 500M, 600M, 1G**, 2G, 3G, 4G, 5G, 6G, 10G, 20G***, 30G***, 40G***,50G****, 60G**** and 100G****
Rate-limiting method	Ingress Policing per circuit (Rate-limiting based on Layer2: (Bit rate including Ethernet frame payload, header, FCS)
Resiliency	Protected and unprotected options (Availability of resiliency option varies by geography)
Automatic protection time	Less than 50msec and Sub-100msec (+Propagation delay) for international circuits
Max frame size *	9,100byte (Ethernet frames including header, FCS and VLAN) Note: Max frame size for Singapore Metro for bandwidths 600Mbps and below is 1990bytes and bandwidths 1Gbps and above is 9100bytes. (Jumbo frame of 9100 bytes can be customized for ≤600Mbps with additional charge)
Link loss forwarding	Standard support – On customer's request
Performance reporting	Supported upon customer's request (with charge)

Table 6. Service Specifications (Ethernet Line)

* : Not available in Dubai – double-check with Regulatory team.
** : Max bandwidth available for international circuits to Dubai and South-Korea.
***: Not available for intra-USA circuits, Turkey and international circuits except intra-Europe, intra-Japan, Singapore and Hong-Kong circuits between 2 Key DCs.
****: Only available for circuits in several cities in Europe between 2 Key DCs (see next paragraph).

7.2.2 Details about Ethernet Line circuits > 40G

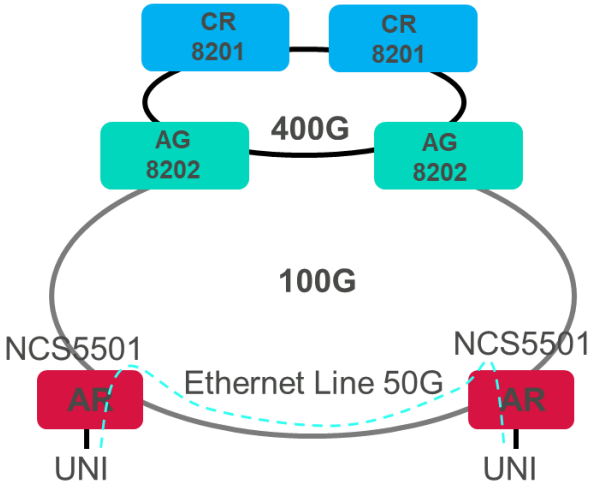
IQNet backbone upgrade to 400G in Europe allowed us to introduce high-speed Ethernet Line services (50G, 60G and 100G) between Key DCs as long as the 2 end-points are located in one of these 18 cities: **UK** (London), **Belgium** (Brussels), **Netherlands** (Amsterdam), **Germany** (Berlin, Dusseldorf, Frankfurt, Hamburg, Hannover, Munich, Stuttgart), **France** (Paris, Marseille, Lyon), **Switzerland** (Zurich), **Spain** (Madrid, Barcelona) and **Italy** (Milan, Turin)

Please**, note** that only **London, Brussels, Amsterdam, Dusseldorf and Frankfurt** use the **Productised** journey in CPQ**.** For the **other enabled cities**, use the **Container** journey.

Depending on the selected bandwidth, the deployment rules are not the same as 50G and 60G circuits will use a shared infrastructure while 100G circuits will use dedicated elements.

The following diagrams describe all the possible scenarios:

1. **Ethernet Line 50G/60G metro services between Key DCs:**



Protected Ethernet Line 50G Metro circuit

In this scenario, shared AR will be used to deliver 50G or 60G services (same as 10G, 20G, 30G or 40G services). As these services will always use shared rings, only Protected services can be delivered.

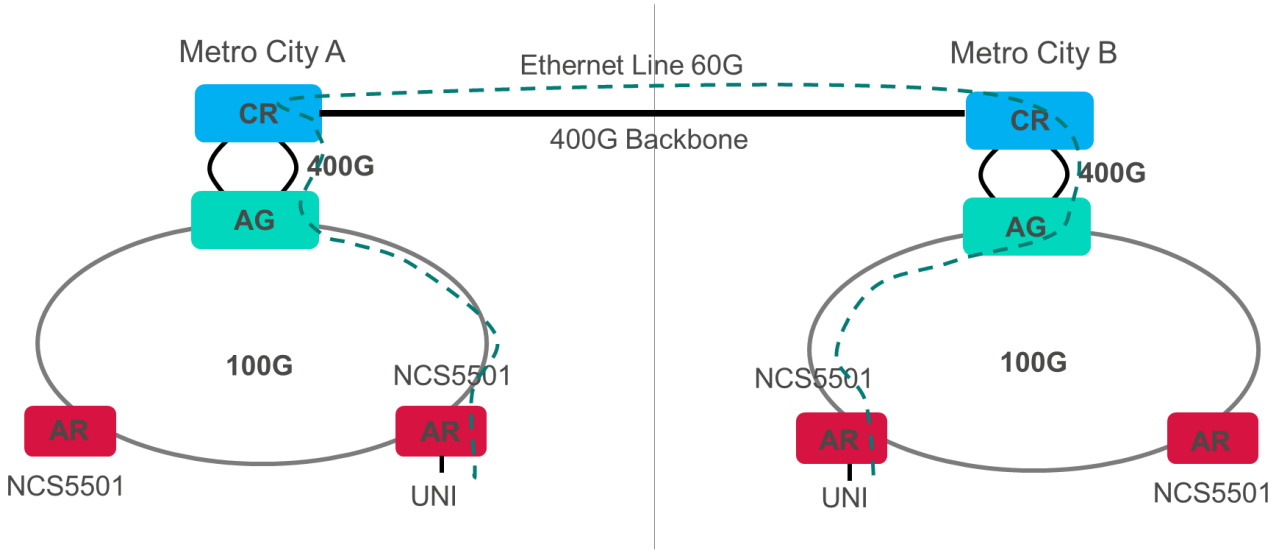
Also as the circuits will use shared rings with termination on Cisco NCS5501 AR, a close monitoring is required from Network planning to determine when additional 100G ring is needed.

From a delivery lead time perspective, there are two cases:

- If there is enough capacity (no additional AR or no new 100G ring), the standard lead time applies.
- If there is not enough capacity, then the extended lead time will apply ie 67 working days.

This check is done during the CST phase.

2. Ethernet Line 50G/60G national/international services between Key DCs



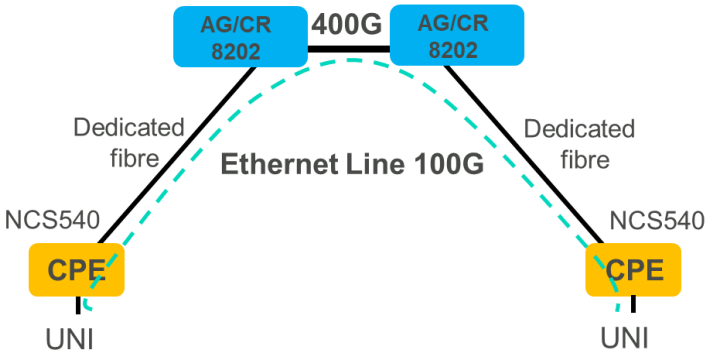
Protected Ethernet Line 60G National (or International) circuit

This scenario is only possible between the Key DCs of the listed cities above for international circuits and the only one national circuit being between Dusseldorf and Frankfurt.

Regarding the other points, the same comments as for the Metro services apply:

- Only Protected services available.
- Close monitoring of the rings to always ensure that there's sufficient capacity
- Standard delivery lead time apply if no capacity upgrade is required otherwise, extended lead time of 67 working days
- Capacity checks done during the CST phase

3. Ethernet Line 100G metro services between Key DCs



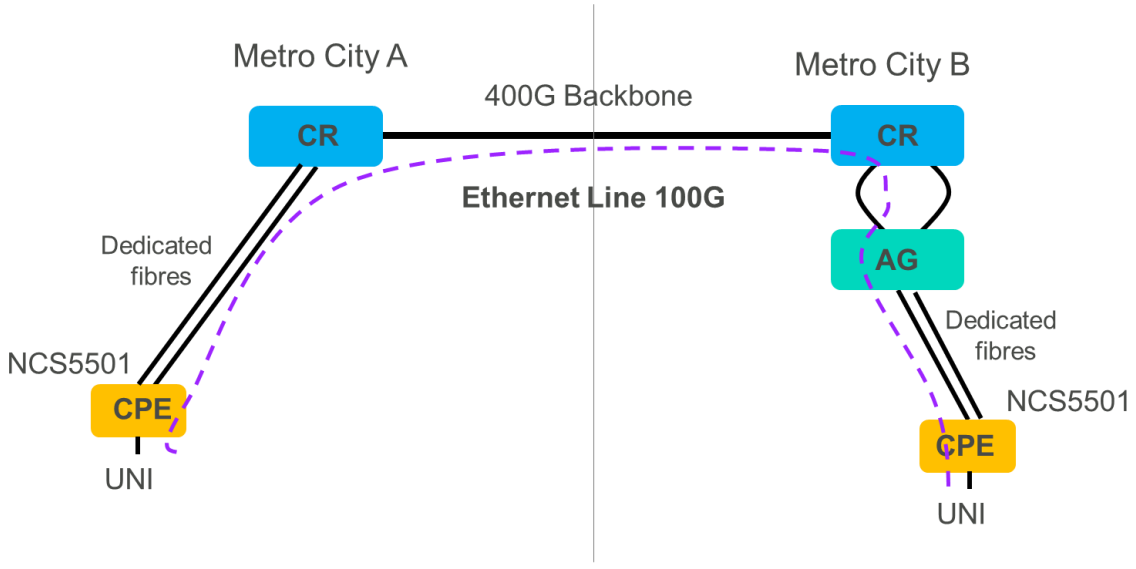
Unprotected Ethernet Line 100G Metro circuit

100G Metro circuits are available in the 5 cities listed above. This solution requires to use dedicated CPEs (NCS540 or NCS5501) and a dedicated fibre to connect the CPE to the AG or the CR router as the Access layer links cannot support 100G circuits.

This service can be delivered either as Unprotected using the NCS540 (Tortin model) as a CPE or Protected with the NCS5501.

From the delivery lead time perspective, the extended lead time always applies (67 working days) as each circuit require a Network build including the CPE installation and the use of a dedicated fibre. Each CPE is dedicated to a customer and is installed in Colt's rack.

4. Ethernet Line 100G national/international services



Protected Ethernet Line 100G National (or International) circuit

Same comments as in the previous scenario apply:

- Service available between Key DCs from the listed cities above
- Dedicated access fibre and dedicated CPEs used
- Available as Unprotected (NCS540) or Protected (NCS5501)
- Extended lead time always apply ie 67 working days.

7.3 Private Ethernet

Ethernet over Fibre is used to deliver the high-speed Private Ethernet service at native Ethernet speeds of 1Gbps, 2Gbps, 3Gbps, 4Gbps, 5Gbps, 6Gbps, 10Gbps, 20Gbps, 30Gbps, 40Gbps, 50Gbps, 60Gbps and 100Gpbs. This service does not use the platforms described below but dedicated Colt's fibre for the highest levels of security and is available in an unprotected or protected configuration. It's available only in the metro areas of Colt's network.

Colt Private Ethernet Services are provided on a physical port (EPL) based on a simple service to port mapping.

Port – based services are transparent to customer VLANs or Ethernet frames with headers containing multiple VLAN tags. Either untagged or tagged frames may be transmitted over such links.

Private Ethernet is built on a Colt's dedicated fibre between two customer sites in the same city which ensure the maximum security and performance. Private Ethernet is only available in cities where MAN fibre infrastructure is present in Europe.



Figure 10. Private Ethernet

7.3.1 Private Ethernet Specifications

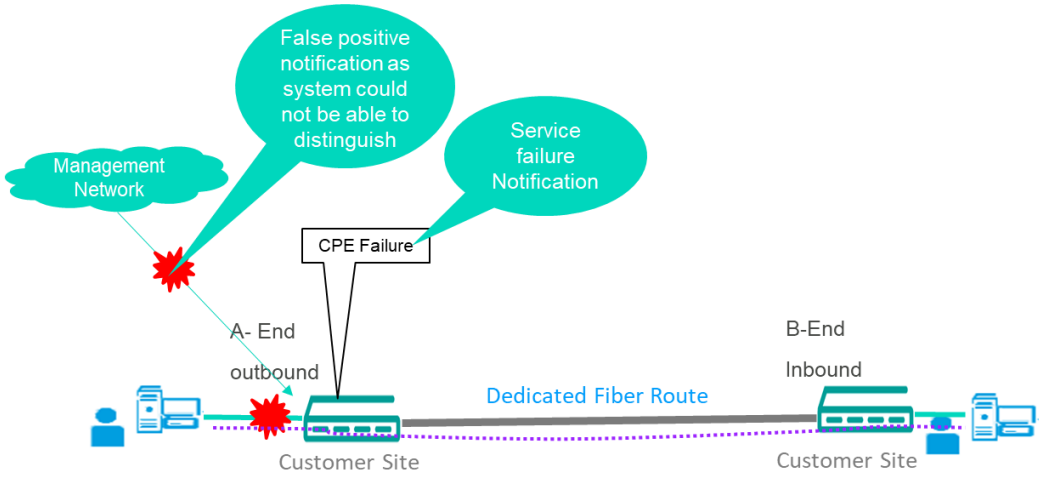
Item	Specifications
Area	Metro (only available in Europe in Colt's metro areas)
Service topology	Point-to-point
Service Type	Guaranteed
Service mapping	Map UNI ports to circuits
Interface *	1000BASE-SX/LX/T 10GBASE-SR/LR/ER 100GBASE-LR4
Bandwidth	1G, 2G, 3G, 4G, 5G, 6G, 10G, 20G, 30G, 40G, 50G, 60G and 100G
Rate-limiting method	Ingress Policing per circuit (Rate-limiting based on Layer2: (Bit rate including Ethernet frame payload, header, FCS)
Resiliency	unprotected by default and protected in option
Automatic protection time	Less than 50msec
Max frame size *	9,100byte (Ethernet frames including header, FCS and VLAN)
Link loss forwarding	Standard support – On customer's request
Performance reporting	Supported upon customer's request (with charge)

Table 7. Service Specifications (Private Ethernet)

* : Interface and frame size specificities are described below in the document.

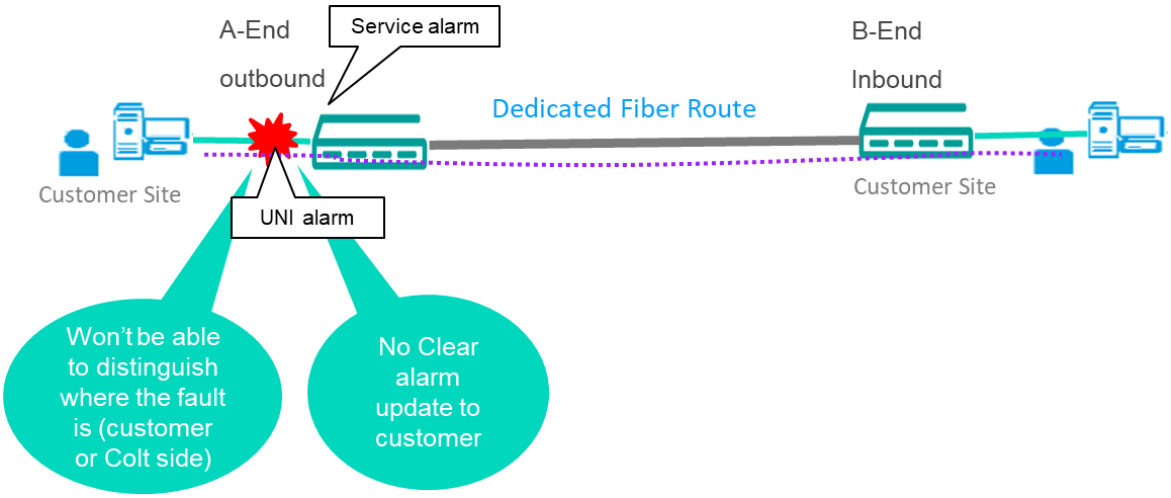
Private Ethernet AMN limitations for above 10G circuits:

1) Out of management issues incorrectly notified:



Private Ethernet is managed via an out-of-band connection as the service is delivered without using a shared platform (IQNet or MMSP) : there is only one management breakout to Private Ethernet management service and this management connection is created in a standard fashion (ie Netcracker) while the Private Ethernet service is “manually” created. Then service will be marked as down if the management service or the interface connecting to the device is operationally down. This may send a false service down notification to the customer if it’s only the management circuit that is down but not the service itself.

2) UNI port down or service down :



When the UNI port goes down (it could be the customer who shuts down the port), the

device generates a service alarm and an information alarm (UNI alarm) to Smart. Then depending on the location, there is 2 different behaviours :

- In Europe : both alarms (UNI and service) are ticketed and notifications of both alarms are sent to the customer.
- In Asia : only the service alarm is ticketed (UNI alarm is suppressed) and service down notification is sent to the customer.

Either way, the customer receives a service down notification which may be a wrong information if the customer himself has shut down the UNI port.

7.4 Ethernet Hub & Spoke

Ethernet Hub and Spoke is compliant with the MEF E-Line definition of an Ethernet Virtual Private Line (EVPL) service and is constructed from several point-to-point EVCs with a single multiplexed Ethernet User Network Interface (UNI) at the hub site.

The Ethernet Hub & Spoke service is a specific case of a VLAN-based service. The service can be configured in one of two modes: a ‘transparent’ mode whereby customer assigned VLANs in frames submitted to the spoke will be preserved and presented transparently at the Hub interface, or a ‘Colt tagging’ mode (or ‘Translation’ mode) whereby Colt will apply an outer VLAN tag to traffic coming from the spoke at the Hub egress interface. The ‘Translation’ mode is the default mode for Hub and Spoke services used by the majority of our customers.

In the ‘transparent’ mode, Spoke sites accept tagged frames with the specified VLAN ID and Ethertype from the customer. In rare cases a single spoke circuit can accept multiple VLAN IDs which is called ‘Multi-VLAN’ spoke, see the paragraph below for more details about “Multi-VLAN”. Customers apply these tags and identify them to Colt at time of order. Frames that are carrying multiple levels of tag (‘double tagged’ packets) are also transmitted and the Colt service will only use the outer tag.

In ‘Colt tagging’ mode or ‘Translation’ mode, Spoke sites may accept untagged or tagged frames from the customer. Colt applies an outer VLAN tag to any frame coming from the spoke at the Hub site egress, which is used to identify the Spoke at the aggregated Hub interface. In reverse, the outer VLAN is stripped at the Hub ingress and the packet is forwarded to the Spoke. Customers define the tag for each spoke at the time of order.

Ethernet Hub and Spoke services in the ‘Colt tagging’ mode may be ordered with one of two standard VLAN Ethertype options:

- The IEEE standard 802.1Q VLAN Tagging
- The IEEE standard 802.1ad VLAN Tagging

Both options support transparency of VLANs contained within the Ethernet frame header.

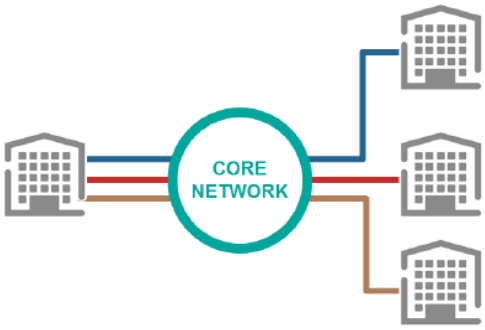


Figure 11. Ethernet Hub & Spoke

7.4.1 Detail on VLAN operation

Ethernet Hub and Spoke uses VLAN tagging to segregate traffic. For ingress traffic at the hub site, the unique VLAN tag ID determines the spoke to which traffic is routed. The Ethernet Hub and Spoke service supports 50 spokes for 1G/10G Hubs and 80 spokes for 100G Hubs with an allowed VLAN ID range of 2 to 4089.

The following diagram shows the Ethernet Hub and Spoke technical architecture.

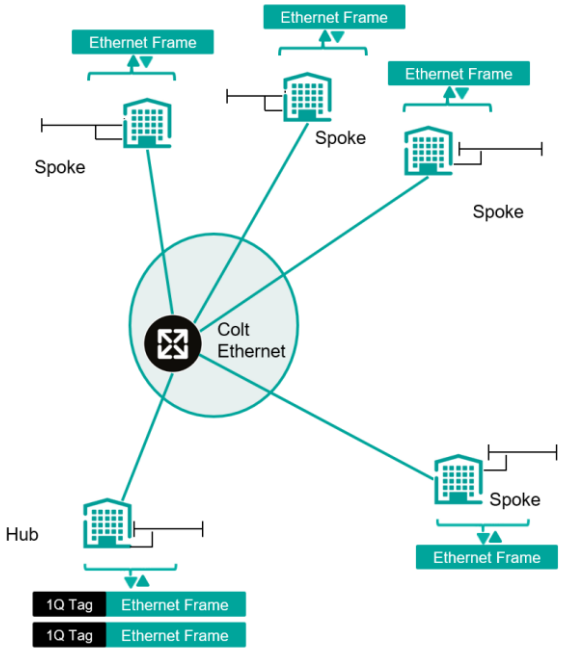


Figure 12. Ethernet Hub and Spoke technical architecture

Colt also supports double tagging by default on the Hub and Spoke service. This allows the service to be transparent to individual customer VLANs on a single spoke. This means that

the Colt Hub &Spoke service is always transparent to additional inner customer VLAN tags.

The mode of operation is the same as for standard single tagging whereby Ethernet frames entering the network at the Hub can be double tagged or not. The Hub will identify the Spoke only based on the outer most VLAN tag, no matter if the frame is single or double-tagged. The same applies for the Spoke ingress in case of “Transparent” mode.

Multi VLAN Spokes

Note: ! This feature is not supported anymore even for customers having Hubs on legacy platforms (NSN / MMSP) !

See the alternatives described further down in this document.

This feature allowed spokes to be designated as multi-VLAN if requested by the customer. In this case the customer defined the individual VLAN tag IDs assigned to the spoke and was also responsible for ensuring that the VLAN tags were added on ingress. They were passed transparently to the Hub. The number of spokes and the overall allowable VLAN ID range is as outlined in section above.

For services delivered either on the IQNet platform or on IQNet and M-MSP (using an I-NNI):

Multi-VLAN spokes are not supported in this case because of limitations on Cisco 5501 : Cisco supports only 32 different ranges on the device, few of these ranges are reserved for internal purpose. Furthermore, Multi-VLAN configuration would cause issues with M-MSP based endpoints (mixed circuits) and some OLO configurations.

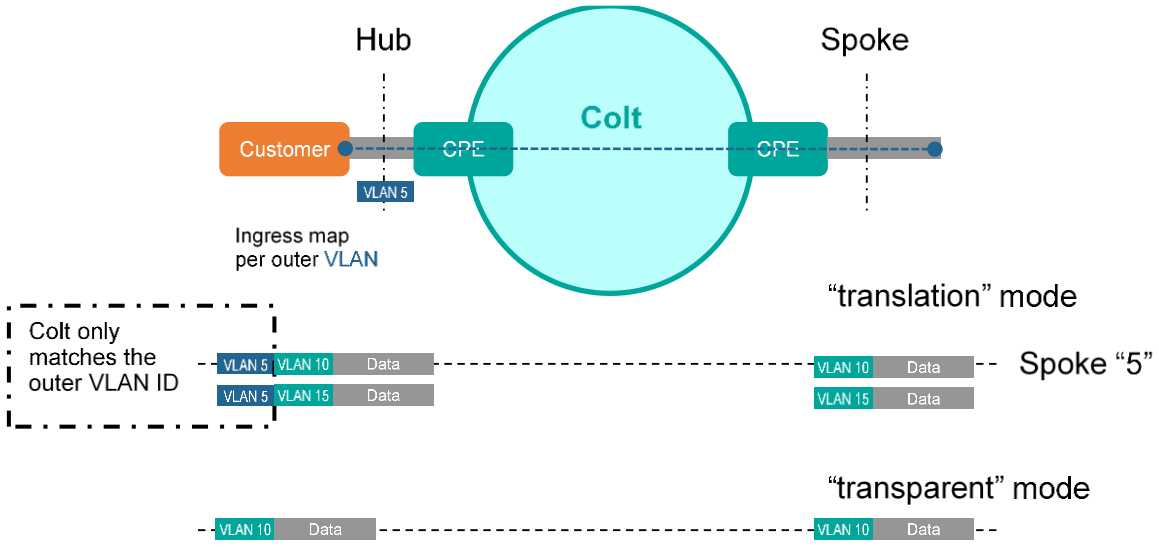
Therefore, high bandwidth Hub with high number of spokes cannot support Multi VLAN scenario due to this restriction.

Recommended alternative solution for customers asking for Multi-VLAN options:

If a customer wants to use multiple VLAN IDs per spoke, the better solution than using the Multi-VLAN configuration (which is not supported on IQnet) is to configure the solution with a double tagged Hub. To use this approach the customer only needs to change the configuration at his own Hub site from single tagged to double tagged, the Spoke sites can remain as single tagged.

The multiple VLAN IDs from the Spoke will all be delivered at the Hub with an extra outer Spoke VLAN tag, resulting in double tagged frames. This allows the customer to use any number of VLAN tags at the Spoke without even specifying them to Colt since they are forwarded transparently. (See the diagram below – VLANs 10 and 15 are transparent in this case and the customer can add as many VLANs as he wants to.)

It is even possible to use the same VLAN IDs at different Spoke sites (e.g. for the same customer applications) since the Hub will distinguish them by the unique outer VLAN ID for each Spoke.



7.4.2 Ethernet Hub & Spoke Specifications

Item		Specifications
Area		Metro*, National* and International
Service topology		Hub and Spoke
Service type		Guaranteed
Hub (Host) Interface		1000BASE-SX/LX/T 10GBASE-SR/LR/ER 100GBASE-LR4**
Spoke (Channel)	**Interface**	100BASE-T 1000BASE-SX/LX/T 10GBASE-SR/LR/ER
	Bandwidth	2M, 4M, 6M, 8M, 10M, 20M, 30M, 40M, 50M, 60M, 100M, 200M, 300M, 400M, 500M, 600M, 1G***, 2G, 3G, 4G, 5G, 6G, 10G
Rate-limiting method		Ingress Policing per circuit (Rate-limiting based on Layer2: (Bit rate including Ethernet frame payload, header, FCS)
Resiliency		Protected and unprotected options (Availability of resiliency option varies by geography)
Automatic protection time		Less than 50msec and Sub-100msec (+Propagation delay)
Max frame size *		9,100byte (Ethernet frames including header, FCS and VLAN) Note: Max frame size for Singapore Metro for bandwidths 600Mbps and below is 1990bytes and bandwidths 1Gbps and above is 9100bytes. (Jumbo frame of 9100 bytes can be customized for ≤600Mbps with additional charge)
Performance reporting		Supported for M-MSP based circuits (with charge)

Table 8. Service Specifications (Ethernet Hub & Spoke)

* : Not available in Dubai – double-check with Regulatory team.
** : A capacity check is required to install a 100G Hub.
*** : Max bandwidth available for international circuits to Dubai and South-Korea.

Spoke limits per hub :

Note that the number of Spokes supported by a Hub varies depending on the hub port speed :

- 1G/10G Hub : limited to 50 spokes.
- 100G Hub : limited to 200 spokes.

For 'high-scale' hubs that have to support more spokes than defined above, it's possible to raise a Capacity check to the Packet Network Planning team in order to track these high demanding hubs.

Hubs and Spoke bandwidth limit and Hub overbooking:

Spoke bandwidth limit:

Depending on where the Hub sits, it's important to note that it will limit the maximum bandwidth of the Spoke. So, refer to the table below to figure out what is the maximum bandwidth that a Spoke site can support:

Hub BW	Allowed spoke without capacity check
1G	600M
10G	1G
100G	10G

For other bandwidth, capacity check is needed to validate the network feasibility.

Hub overbooking:

Note that large spoke bandwidths do consume the available Hub bandwidth and it will reduce the overall hub capacity available and therefore the total number of spokes that can be provisioned.

It is possible to overbook the Hub on customer's request but the following should be noted:

- Maximum overbooking is 5 x Hub bandwidth e.g. 50Gbps for a 10Gbps Hub **(strict limit)**
- It should be clearly explained to the customer that the traffic management of the Hub is his responsibility : if the actual utilisation of the Hub port exceeds the port speed, the customer may experience service degradation.
- Performance SLA (Jitter, Frame loss and Latency) become invalid if a fault is reported as a result of the Hub over utilisation : this should be clearly mentioned and agreed with the customer and a special annotation added with the order. In such case, you can use the following liner : *"As the hub is oversubscribed customer might face degradation on services if the actual utilisation exceeds the port speed. Performance SLA becomes invalid if a fault is reported as a result of over utilisation."*

In Asia, Hub overbooking is only supported on IQNet Hubs. Hub overbooking is not available for M-MSP Hubs.

EFM access : (ULL copper)

EFM access is not supported on IQNet Hubs. This is a legacy technology that has not been tested on the new IQNet platform and then should not be used as an access for new services.

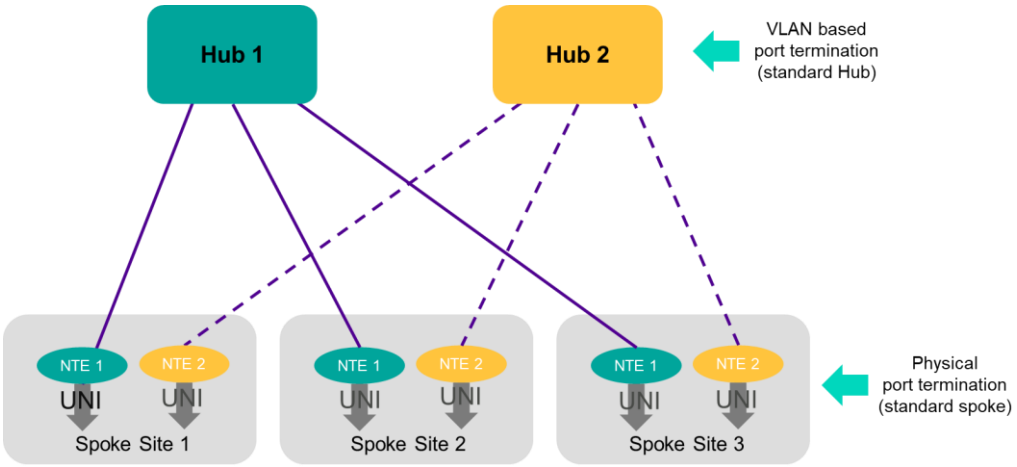
However, there are some scenarios where old hubs are migrated to the new IQNet platform and then the spokes must be shifted as well but with a new access technology as there is no workaround.

Don't forget to raise this point to the customer awareness about this dependency whenever needed.

7.4.3 Dual Ethernet Hub & Spoke scenarios

Some customers sometimes require to set up a dual Hub&Spoke configuration for their network. It's possible to use one the 3 following scenarios depending on the customer's needs :

1) Dual Hub, dual Spoke, dual handoff (separate NTE)

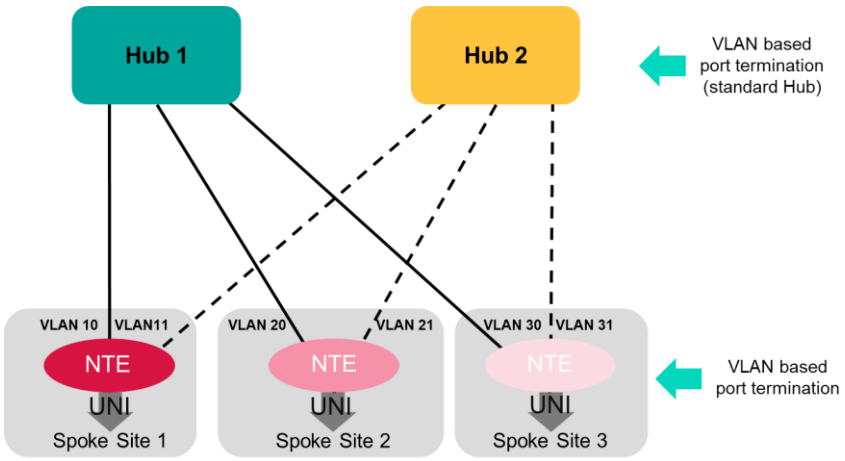


This is a standard solution: the traffic coming from each hub is delivered on the NTE UNI (User Network Interface) to that it's connected.

In this scenario, the customer must order 2 x hubs and 2 x spokes. The 2 x spokes being located in the same building.

Please note that the customer needs to order the Diversity option if he needs diversity between the Spokes installed in the same location.

2) Dual Hub, dual Spoke, single handoff (same NTE)

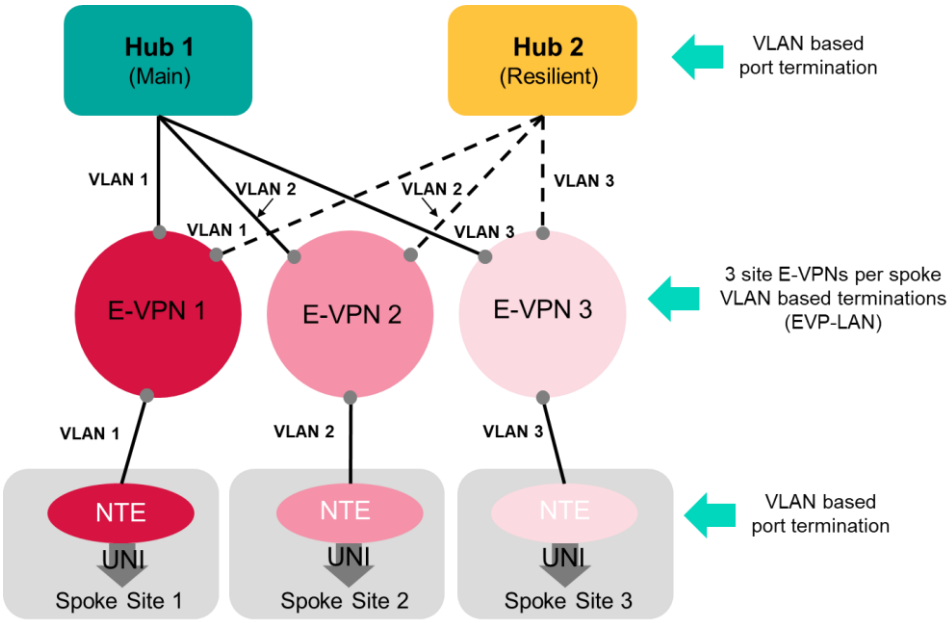


This is not a Standard solution as additional details are required at the Spoke end: the traffic coming from each hub (Hub 1 and Hub 2) is delivered on the same NTE UNI.
Therefore to segregate the traffic coming from the hubs, it's necessary to tag it in the NTE so it's delivered with two different VLAN/EVPL's (for example VLANs 10 and 11 in the Spoke Site 1 of the diagram) on the UNI.

The VLAN operation on Hub and spoke site will be transparent as per standard scenario of tagged spoke site.

A SE should be engaged to raise a SR to capture the VLAN termination details on the spoke end. Pre-approved SR_20821 to be followed. You can get in touch with the Solution Design team to get more information about it.

3) Resilient Hub & Spoke using E-VPN based spokes



This is a bespoke solution that has been delivered a couple of times. The solution allows to have one spoke end attached to two Hubs with an automatic failover from Hub1 to Hub2 if Hub1 fails. Note that during normal operations, just the Hub1 should be active: the customer has to implement some form of failover mechanism such as HSRP/VRRP to ensure that there is a main Hub and a failover Hub.

In this scenario, each Spoke is configured in a 3 sites VPN made up of the spoke and the 2 Hubs. For example, E-VPN 1 includes Hub 1, Hub 2 and Spoke site 1 and so on…

To segregate the incoming traffic from the customer at the hubs, it's necessary to use a VLAN so the Hub knows to which E-VPN the traffic must be forwarded. So if the Hub 1 receives traffic from the customer tagged with the VLAN 1, he "knows" this traffic must be sent in the E-VPN 1. The same for VLAN 2 and VLAN 3.
At the Spoke end, the UNI port can be configured as a VLAN based port or a physical port depending on the customer requirements.

Note that the use of multiple VPN is to prevent direct inter-spoke communications without going via the Hub site.

This solution is bespoke, a SE needs to be engaged to raise a SR but as this solution has been already implemented, previous SR can be reused as reference (for example the SR_20821can be reused – you can get in touch with the Solution Design team to get more information about it).

7.5 Ethernet VPN (E-VPN)

7.5.1 Positioning

Ethernet VPN (Ethernet VPN) is an integral part of the overall Ethernet portfolio of services and provides a multipoint, any-to-any switched service to customers. This service is available on IQNet and the Modular MSP platforms. Ethernet VPN is an alternative to an unmanaged IPVPN where the ease of management and bandwidth flexibility will appeal to customers looking to either manage the IP layer in-house using existing expertise or to add their own services to the Colt layer 2 product.

7.5.2 Summary

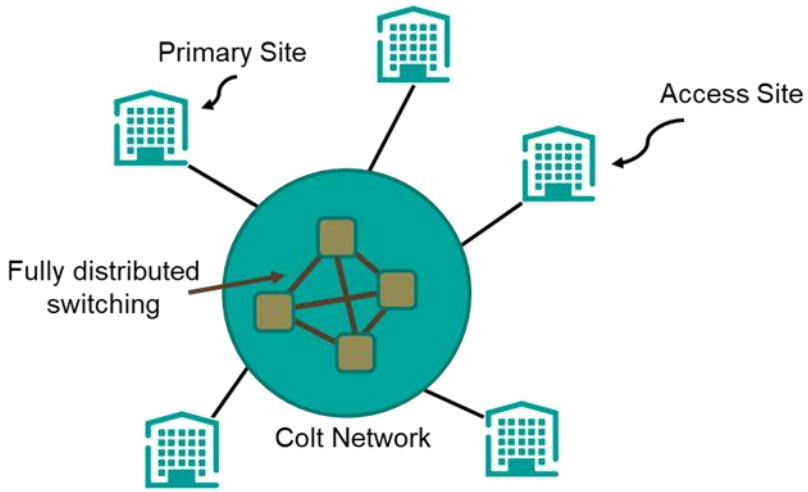
The Ethernet VPN service is a Layer 2 service offering customers the ability to connect their sites together in an any-to-any logical topology, allowing each site to communicate with every other site. Multipoint connectivity is achieved via a fully distributed switching function that is implemented in the Colt network. This allows any Ethernet frame entering the network at one site to be switched and delivered to any destination site. The service offers the following:

- Any-to-any switched service

- Access speeds from 2Mbps through to 40Gbps
- Service dedicated to a single customer, no bandwidth contention
- Metro, National and International coverage with Off-Net access also available

Ethernet VPN is compliant to the MEF E-LAN definition of an Ethernet Private LAN (EP-LAN) and is a multipoint-to-multipoint Ethernet Virtual Connection (EVC). The service is fully transparent to Layer 2 protocols (802.1Q VLANs, spanning tree, etc)

7.5.3 Topology



An Ethernet VPN consists of the following components:

- **Ethernet VPN Primary Site –** Identified by the customer and captured on the order form. The location of the primary site performs an operational function - it indicates to the Colt Operations teams where to set up internal monitoring circuits (SOAM). These circuits are configured from each Access Site to the Primary Site.
- **Ethernet VPN Access Site –** Ethernet VPN access sites represent all the other customer sites on the multi-site network.

The customer location addresses of the Primary Site and Access sites are captured on the Order Form.

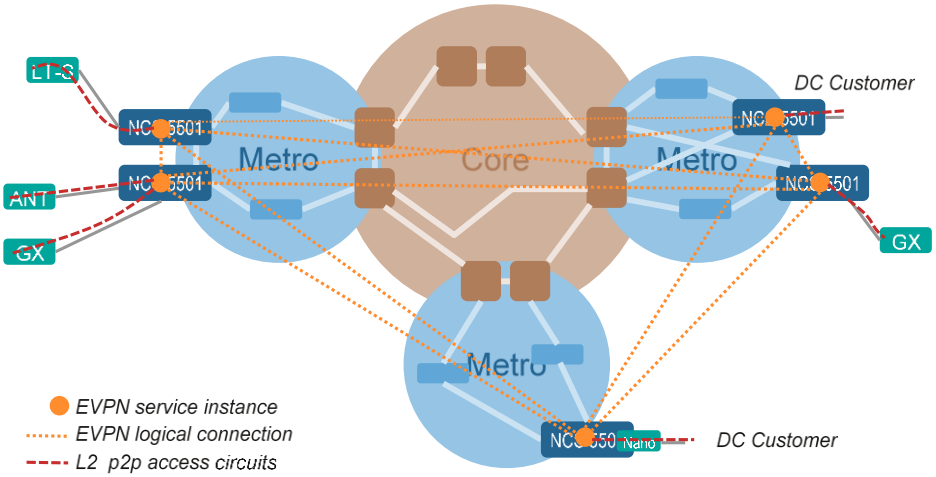
The switching function is fully distributed throughout the Colt Network with each core device acting as a Layer 2 switch. This ensures the lowest latency for traffic between any two sites.

7.5.4 Technical Aspects of the E-VPN

Within the IQNet core network (Cisco metro and core) Colt will use the IETF EVPN standard to implement the Ethernet VPN services. In this context “EVPN” refers to the MPLS/BGP

based service as specified by the IETF (RFC7432 etc.) and implemented on the IQNet Cisco devices. This should not be confused with the Colt Ethernet VPN (E-VPN) service itself. The EVPN service is from the same IETF standard group as the point-to-point EVPN VPWS service introduced in Drop 1 of IQNet (Ethernet Line). For the multipoint support it provides the Ethernet MAC address learning via the MP-BGP protocol and enables multiple sites to be connected to one logical Ethernet L2 bridge domain.

The BGP EVPN service will create a logical full mesh between all Cisco NCS 5000 routers which connect to a service endpoint either directly, via star access or via an NNI (E-NNI, I-NNI or DCA NNI).

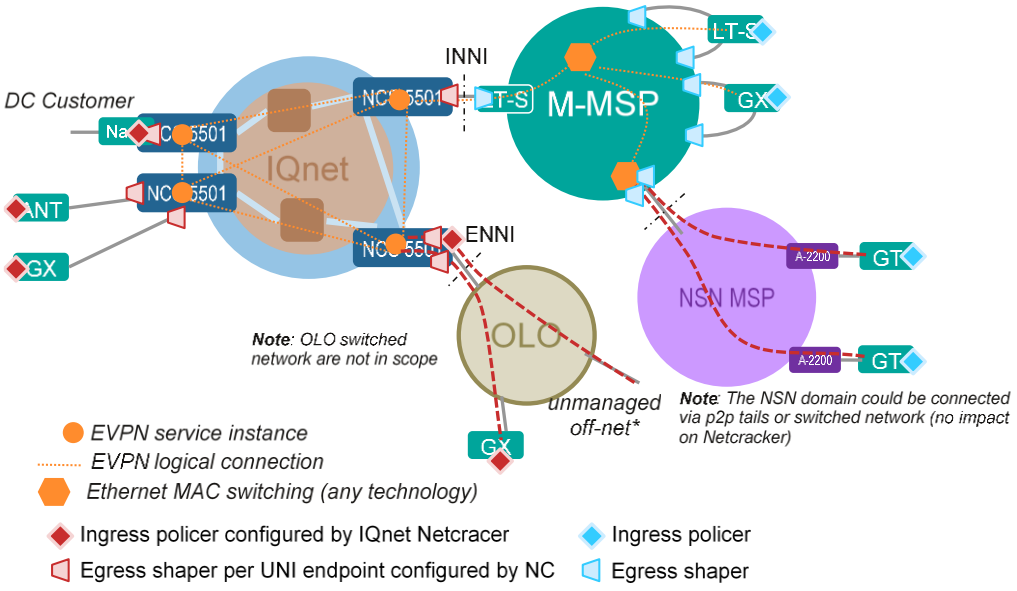


7.5.5 Multi-domain Ethernet VPNs

Multi-domain E-VPNs allow multi-site networks to be created where sites are distributed across IQNet, M-MSP and even NSN platform locations. This enables the following extended E-VPN combinations to be built:

- Existing E-VPN networks that have been configured and built on M-MSP can be extended by using an E-VPN domain on IQNet
- When building E-VPNs on IQNet, sites where IQNet is not yet available can be connected using M-MSP or even NSN if absolutely necessary

In order to create these extended multi-site networks MAC switching on multiple network domains is used. This multi-domain switching allows for the expansion IQNet based Ethernet VPN services to M-MSP (including NSN and OLO via M-MSP) or adding new endpoints on IQNet to existing M-MSP based E-VPN services.



The interconnection between an IQNet E-VPN domain and MSP E-VPN domain is achieved via an Internal NNI (I-NNI).

Access to off-net sites via 3rd party OLO services is achieved via External NNIs (E-NNI).

7.5.6 E-VPN Access Specifications

Item	Specifications
Area	Metro*, National* and International
Service topology	Any to Any
Service Type	Guaranteed
Service mapping	Map UNI ports to circuits
Interface	100BASE-T 1000BASE-SX/LX/T 10GBASE-SR/LR/ER 100GBASE-LR4
Bandwidth	2M, 4M, 6M, 8M, 10M, 20M, 30M, 40M, 50M, 60M, 100M, 200M, 300M, 400M, 500M, 600M, 1G**, 2G, 3G, 4G, 5G, 6G, 10G, 20G***, 30G***, 40G***
Rate-limiting method	Ingress Policing per circuit (Rate-limiting based on Layer2: Bit rate including Ethernet frame payload, header, FCS)
Resiliency	Protected and unprotected options
Automatic protection time	Less than 50msec and Sub-100msec (+Propagation delay) for international circuits
Max frame size *	9,100byte (Ethernet frames including header, FCS and VLAN) *Note: Max frame size for Singapore Metro for bandwidths 600Mbps and below is 1990bytes and bandwidths 1Gbps and above is 9100bytes*
Link loss forwarding	Not applicable
Performance reporting	Supported for M-MSP based E-VPNs only

Table 9. Service Specifications (Ethernet VPN)

* : Not available in Dubai – double-check with Regulatory team.

** : Max bandwidth available for international circuits to Dubai and South-Korea.

** : Bandwidths >10Gbps only available for on-net sites in Europe, Japan, HK and Singapore

7.5.7 MAC Address Limitations

Ethernet VPN 'any to any' functionality is based on Layer 2 MAC switching and an associated MAC Address Table is created for each independent E-VPN instance. The IQNet core devices (Cisco NCS 5501) have finite MAC table capacity and this is shared between the multiple E-VPN instances that will be configured on that device. Therefore each E-VPN is set up initially with a limit of **2000 MAC addresses as default**. This is normally sufficient to support the vast majority of E-VPN configurations. However, if the customer has a particularly large E-VPN or requires additional MAC Address support for other reasons then the MAC Address Table size can be increased.

Any change in the size of the MAC Table from the default will need to be triggered by a special request to the Design/Configuration teams.

7.5.8 Off-Net E-VPN Access sites

Any E-VPN Access Site that is off-net and delivered via a 3rd party OLO service **must have a demarcation device installed**. These devices are mandatory because they are necessary to support the SOAM monitoring circuits that are set up between each E-VPN Access Site and the Primary Site. Installation of OLO demarcation devices is standard in all Colt Countries across Europe, Asia and North America. **For countries outside these i.e. Global Reach countries then a bespoke request for the installation of a demarcation device with the OLO service must be included with the order**. This will ensure that the International Service Delivery team is triggered to raise an order request for the OLO tail including the NTE on the 3rd party organisation that will undertake the installation.

Note: Installation of the demarcation device in Global Reach countries will incur additional cost and extended lead times.

8 Bandwidth & Interface Option

Colt Ethernet Services support a wide range of optical bandwidths and interface standards.

The available bandwidths and interface types are:

Speed	Presentation/Interface	Cable	Connectors
2Mbps to 100Mbps*	10BASE-T** 100BASE-TX	Copper Category 5 or better	RJ45
200Mbps to 1Gbps	1000BASE-T	Copper Category 6 or better	RJ45
	1000BASE-LX	Singlemode 1310nm	LC, SC, FC/PC***
	1000BASE-SX	Multimode 850nm	LC, SC, FC/PC***
2Gbps to 10Gbps	10GBASE-SR	Multimode 850nm	LC, SC, FC/PC***
	10GBASE-LR	Singlemode 1310nm	LC, SC, FC/PC***
	10GBASE-ER	Singlemode 1550nm	LC, SC, FC/PC***
20Gbps to 100Gbps	100GBASE-LR4	Singlemode 1310nm	LC, SC

Table 10. Ethernet Services interface options

* Services delivered between 2Mbps and 100Mbps using Colt's Ethernet over MSP platform or in conjunction with the Colt Ethernet Demarcation Device can also be delivered on 1Gbps interfaces for an additional charge (Not available in Asia).

** 10BASE-T interfaces are not delivered anymore on the new NTE used on the IQNet platform.

*** FC/PC connector is not available in Asia and we are not offering it anymore in Europe/North-America.

8.1 Interface Settings

Colt Ethernet Services provide fully dedicated bandwidth and are full-duplex, protocol-independent services, which means customers can use the full amount of the bandwidth supplied in either direction.

All Ethernet interfaces are compliant with the IEEE 802.3-2015 MDI specification (Medium Dependant Interface), as defined in the UNI specification. In the case of Ethernet presentations, auto-negotiation is enabled according to IEEE 802.3-2015 by default.

The full range of interfaces available at the customer site are shown in the table below:

Presentation Interface	Standard	Fibre type	Regional Availability	Max Distance
10 Base-T[1]	IEEE802.3i	N/A	All	100m including patch panels
100 Base-TX	IEEE802.3u	N/A	All	100m including patch panels
1000BASE-T	IEEE802.3ab	N/A	All	100m including patch panels
1000BASE-SX [3]	IEEE802.3z	MM 850nm	All	550m
1000BASE-LX	IEEE802.3z	SM 1310nm	All	5Km

10GBASE-LR (LAN PHY)	IEEE802.3ae	SM 1310nm	All	10Km
10GBASE-ER (LAN PHY)	IEEE802.3ae	SM 1550nm	All	40Km
10GBASE-SR (LAN PHY)	IEEE802.3ae	MM 850nm	All	275m
100GBASE-LR4	IEEE802.3ba	SM 1310nm	All	10Km

Table 11. Ethernet interface options

1 – 10BASE-T interfaces are not provided anymore on the NTE used with the IQNet platform.

In large and structured environments such as data centres, the selected presentation may be conditioned by factors beyond either Colt's or the customer's direct control, such as the distance between the Colt terminating equipment and customer equipment and the specification of intermediate 'meet me' cross-connect cabling facilities.

8.2 Copper and Fibre Interface Options

Connector type	Regional Availability
RJ45	All
LC-Duplex	All
SC-Duplex	All
FC/PC*	Europe and the USA
MPO/MTP	Europe

Table 12. Connector Type Availability

* Not offered anymore with the new NTE (IQNet platform)

8.3 Copper and Fibre Interface Specifications

Connector types	Fibre Type	Cable
RJ45	N/A	Cat 5e or above
LC-Duplex, SC-Duplex, FC/PC*	MultiMode 850nm	50µ or 62.5µ multi mode fibre (MMF)
LC-Duplex, SC-Duplex, FC/PC*	SingleMode 1310nm	10µ single mode fibre (SMF)
LC-Duplex, SC-Duplex, FC/PC*	SingleMode 1550nm	10µ single mode fibre (SMF)

Table 13. Interface Specifications

* Not offered anymore with the new NTE (IQNet platform)

8.3.1 RJ45 Connector

The RJ45 connector is specified in IEC 60603-7. It is the standard connector for Class D and E (Category 5 and 6) structured cabling. For all services shielded RJ45 connectors are used.

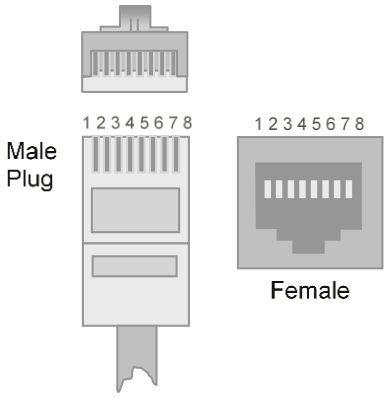


Figure 13. RJ45 Connector

Contact	10BASE-T* and 100BASE-TX		1000BASE-TX	
	MDI signal	MDI-X signal	MDI signal	MDI-X signal
1	TD+	RD+	BI_DA+	BI_DB+
2	TD–	RD–	BI_DA–	BI_DB–
3	RD+	TD+	BI_DB+	BI_DA+
4	Not used	Not used	BI_DC+	BI_DD+
5	Not used	Not used	BI_DC–	BI_DD–
6	RD–	TD–	BI_DB–	BI_DA–
7	Not used	Not used	BI_DD+	BI_DC+
8	Not used	Not used	BI_DD–	BI_DC–

Table 14. RJ45 Pin Assignment

*10BASE-T interfaces are not delivered anymore on the new NTE used on the IQNet platform.

8.3.2 Duplex SC Connector

The duplex SC connector is specified in IEC 61754-4:

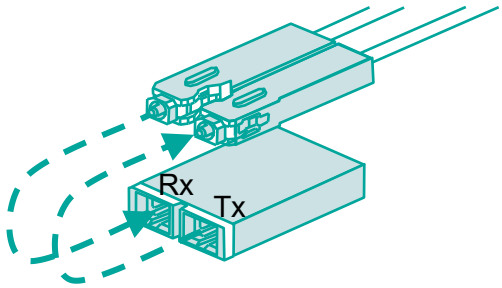


Figure 14. Duplex SC Connector

8.3.3 Duplex LC Connector

The duplex LC connector is specified in EIA/TIA-604-10.

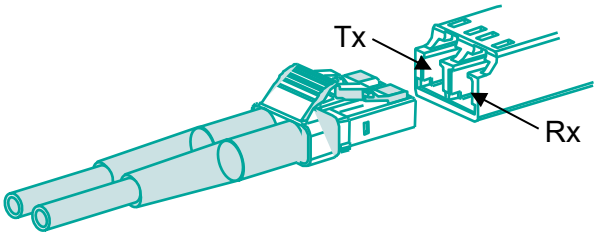


Figure 15. Duplex LC connector

8.3.4 FC/PC connector

The FC connector has been standardized in a FOCIS 4 (Fibre Optic Connector Intermateability Standards) in EIA/TIA-604-04. The PC stands for the polish option “Physical Contact”. This connector is only available as simplex, so it needs to be ensured that Rx and Tx match.

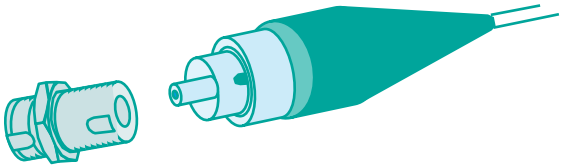


Figure 16. FC/PC connector

These connectors are not delivered anymore on the new NTE (IQNet platform).

8.4 Ethernet MTU

The Ethernet MTU is relevant to services that are provided on Ethernet-framed interfaces . The following table shows the minimum supported Maximum Transmission Unit (MTU) size in bytes for Colt’s Ethernet services provided with an Ethernet presentation.

Access Type	Interface	MTU	Comments
ULL-EFM	10BaseT* 100BaseTX	1572 *Default*	On-net and Ethernet First Mile delivery. MTU may be higher on some Colt platforms.
Colt Fibre & ULL Fibre	10BaseT* 100BaseTX 1000BaseX 10GbaseX 100GbaseX	9100 ** *Default*	Max frame size for Singapore Metro for bandwidths 600Mbps and below is 1990bytes and bandwidths 1Gbps and above is 9100bytes.
3rd Party Leased Lines	10BaseT* 100BaseTX 1000BaseX	1534	Most of Colt’s partners support larger frames, including jumbo frames (9k) on 1Gbps interfaces. Can be confirmed on request.

Table 15. Ethernet Service MTU by access technology

*10BASE-T interfaces are not delivered anymore on the new NTE used on the IQNet platform.

** IQNet packet platform can support up to 9186 bytes but a case by case check is required before committing more than 9100 bytes to customers.

8.5 Protocol Transparency

Colt Ethernet services are transparent to protocols operating at the higher layers of the 7 layer OSI model. This means customers can run their own VLAN, broadcast and multicast domains across the infrastructure to suit specific requirements.

Ethernet services are transparent to Layer 3 protocols providing a single networking solution for voice, video and data requirements.

About L2 protocols listed in the table below, we tend to be referring to our on-net services and infrastructure. However, it may applies as well to OLO providers as we do mandate/require that they are transparent to the majority of L2CP protocols as well. Therefore, where OLO services are involved, it should be double checked in the OLO provider list.

Parameter	Transparent / Value
802.1Q VLAN Transparency	Port-based – Yes – (except VLAN-0 tag might be removed) VLAN-Based – Yes – if transparent mode or double-tagging is selected
802.1P Class of Service	Yes
Default set port to full-duplex	Yes
Default enable auto-negotiate	Yes
Default disable Flow-control (pause-frames)	Yes
Default set port to MDI or AUTO.	Yes (Electrical only)
MAC learning	E-Line – Disabled E-LAN – Enabled, per VLAN
MAC Address Support	E-Line – Not Applicable E-LAN – 8K
MAC Filtering	No
IEEE L2CPs	
Spanning Tree Protocol (STP)	Yes
Rapid Spanning Tree Protocol (RSTP)	Yes
Multiple Spanning Tree Protocol (MSTP)	Yes
Flow Control - PAUSE (802.3x)	No
Link Aggregation Control Protocol (LACP, LAMP)	Yes – except Ethernet Hub & Spoke and Ethernet VPN (E-LAN)
Port Authentication protocol (802.1x)	Yes
Link Layer Discovery Protocol (LLDP)	Yes
Generic Attribute Registration Protocol (GARP) (so GARP VLAN Registration Protocol GVRP)	Yes

Parameter	Transparent / Value
Cisco Protocols	
Cisco Port Aggregation Protocol (PAgP) Cisco Discovery Protocol (CDP)	Yes
Cisco Uni Directional Link Detection (UDLD)	Yes – except Ethernet VPN (E-LAN)
Cisco VLAN trunking Protocol (VTP)	Yes
Cisco Dynamic Trunking Protocol (DTP)	Yes
Cisco Inter Switch Link (ISL)	Yes
Other Protocols	
Transparent for 802.2 LLC/SNAP (Ethertype < 0x0600)	Yes
0x0800 Internet Protocol, Version 4 (IPv4)	Yes
0x0806 Address Resolution Protocol (ARP)	Yes
0x8035 Reverse Address Resolution Protocol (RARP)	Yes
0x809b AppleTalk (Ethertalk)	Yes
0x80f3 AppleTalk Address Resolution Protocol (AARP)	Yes
0x8100 IEEE 802.1Q-tagged frame	Yes
0x8137 Novell IPX (alt)	Yes
0x8138 Novell	Yes
0x86DD Internet Protocol, Version 6 (IPv6)	Yes
0x8847 MPLS unicast	Yes
0x8848 MPLS multicast	Yes
0x8863 PPPoE Discovery Stage	Yes
0x8864 PPPoE Session Stage	Yes
0x888E EAP over LAN (IEEE 802.1X)	Yes
0x88A2 ATA over Ethernet	Yes
0x88E5 MAC security (IEEE 802.1AE)	Yes
ITU.T G.8032 ERPS R-APS Control Messages	Yes – except Ethernet VPN (E-LAN)

Table 16. Protocol Transparency

Important note about the protocol transparency and the full line rate on IQNet :

Due to Cisco limitations, on IQNET star access, the bandwidth ordered includes VLAN overhead on the access component, hence the usable bandwidth may be slightly less than the bandwidth ordered.

The VLAN overhead is used to ensure LLDP and LACP protocols transparency. If those protocols are used by the customer, the impact on the bandwidth varies according to the size of the frames :

- 64 bytes frames = ~ 4.5% throughput loss
- 1518 bytes frames = ~ 0.3% throughput loss
- 500 bytes frames (average size) = ~ 1% throughput loss.

This may be noticed by customers when doing service testing during handover.

For the moment, there is no workaround to fix this issue.

8.6 Link Loss Forwarding

Link Loss Forwarding (LLF) exist to propagate link failure signals to the service interfaces at each end of a circuit. This enables the customer's client devices (such as routers) to stop sending data over the affected links in a controlled manner and where possible redirect traffic as necessary, so as to support the continued operation of the network under error conditions.

Client Signal Failure (CSF) is often taken to be included in LLF but is in fact a separate function. Whereas LLF provides the function of signalling an internal network connectivity interruption to the UNI on the Colt equipment (User Network Interface), CSF communicates link failure between the UNI and the customer's equipment. LLF and CSF protocols are relevant to services with Ethernet interfaces whilst AIS signals apply to OTN services.

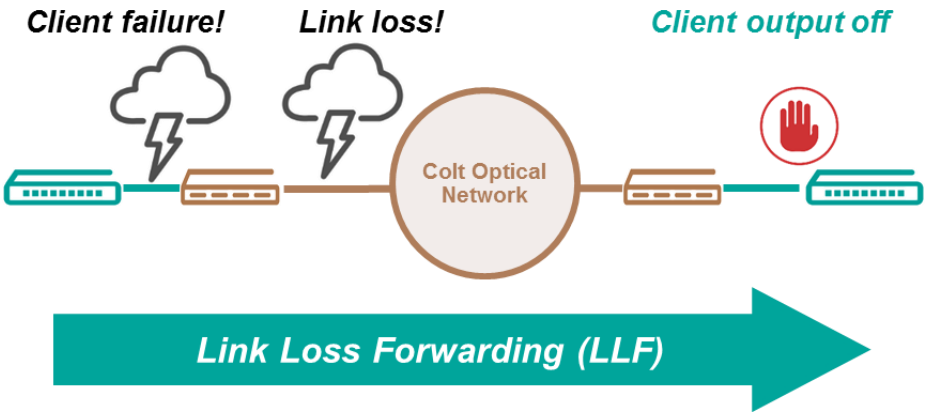


Figure 18. Client Signal Failure & Link Loss Forwarding

Colt's Ethernet Services support LLF, CSF, however a small number of platforms may be an exception.

Link Loss forwarding (LLF) is supported on Colt's Ethernet Line services. Client Signal Failure specifically is only supported on Point-to-Point delivered over Colt Fibre or ULL Fibre. For >10G Private Ethernet services (using the NCS540 device), the LLF/CSF is partially available as network to UNI notifications work while UNI to UNI is not supported.

In the case of services provisioned using EFM or 3rd party access in conjunction with the Ethernet Demarcation Device, there is support for LLF towards the EFM or 3rd party access site.

Access Type	Technology	LLF/CSF
Colt Fibre	Ethernet over IQNet (both ends on the same platform) Ethernet over MSP (both ends on the same platform)	Yes

Access Type	Technology	LLF/CSF
	Ethernet over Fibre	
ULL Fibre (Asia)	Ethernet over IQNet Ethernet over MSP	Yes
3rd Party Leased Line	SDH	No
3rd Party Leased Line	Ethernet (with Colt Ethernet Demarcation Device)	Yes
3rd Party Leased Line	Ethernet (without Colt Ethernet Demarcation Device)	No
Unbundled local loop (ULL)	Ethernet First Mile (EFM)	Yes

Table 17. LLF support

Also, depending on the scenarios, the feature may be or not supported. See the table below:

A-end device	B-end Device	LLF/CSF support
IQNet Accedian GX/LT-S	IQNet Accedian GX/LT-S	Yes
IQNet Cisco NCS 55xx (non-LAG) 10/100G interface only, no 1G	IQNet Cisco NCS 55xx (non-LAG) 10/100G interface only, no 1G	Yes
IQNet Accedian GX/LT-S	M-MSP Accedian GX/LT-S/GT/GT-S	No
IQNet Cisco NCS 55xx (LAG)	IQNet Cisco NCS 55xx (LAG)	No
IQNet Accedian GX/LT-S	IQNet Accedian ANT/Nano	No
IQNet Cisco	IQNet/M-MSP Accedian	No
IQNet/M-MSP	NSN MSP	No

Table 18. LLF support per scenario (platform/devices)

LLF, CSF are **not enabled by default** for Colt's Ethernet Services but can be activated on request. We don't currently have the option to change this setting via the ordering system, so requests to enable LLF/CSF must be submitted using the 'notes fields'.

9 Service Features

9.1 Performance Reporting

Service Performance Reporting is available as a standard feature for Colt Ethernet services. This is offered via Colt Analytics, a reporting tool that enables customers to view the performance of their Colt Ethernet services in near 'real time'. It is based on the Accedian Skylight Software Suite and provides the following features/benefits for customers:

- Ability to view general service characteristics
  - Service Availability, Circuit Bandwidth Utilisation, Port Utilisation
- Detailed Reports on Ethernet Key Performance Indicators (KPIs)
  - Round Trip Delay, Jitter, Frame Loss
- Variable Reporting periods can be selected to allow the viewing of historic data and to focus in on any specific time periods of interest
- Option to download reports – these can be Excel based CSV files and PDFs
- Support for all Ethernet topologies – Ethernet Line, Ethernet H&S, E-VPN and Private Ethernet

Access to the Performance Reporting tool is via Colt Online and therefore customers must have a Colt Online Account set up. They must also have the appropriate Role assigned within Colt Online i.e. *Network Services – Performance Reporting*. This can be done through Self-Registration or via their Customer Relationship Management (CRM) team.

Colt Analytics will provide performance reporting access to the following Ethernet inventory:

- All Ethernet service provisioned on the IQNet platform. This includes hybrid services where one end is delivered on IQNet and the other end is delivered on M-MSP
- All Ethernet services where InfoVista (the previous performance reporting tool) was enabled

All new services delivered on IQNet will have performance reporting automatically enabled and available at no extra charge. It is 'bundled' as part of the basic Ethernet services package.

Where a customer Ethernet service is not visible in the performance reporting tool e.g. because it was delivered on M-MSP. then a Modification (MOD) order will need to be raised in order for the Service Delivery teams to perform the necessary configuration work to enable performance reporting. This may require a Cease and Re-provide and carries a one-off charge (NRC) of €500.

Note: ***It will not be possible to configure performance reporting for Ethernet services that have been provisioned on the older platforms – EoSDH and NSN MSP.***

'End to End' Performance Reporting is also available for off-net services as long as a Colt Demarcation Device has been installed on the end of the OLO tail.

Performance Reporting is available as a service feature globally – Europe, Asia, North America

9.2 Class of Service

Colt's CoS feature provides customers with a means to differentiate their traffic within their own service's bandwidth and therefore to allocate an explicit prioritisation. CoS is therefore relevant only when a customer causes temporary congestion on their Ethernet service by sending more data than the capacity of that Ethernet service.

Five levels of prioritisation can be defined:

- **Premium** - Has the highest priority in case of congestion
- **Business 1** – Has the second highest priority in case of congestion
- **Business 2** – Has the third highest priority in case of congestion
- **Business 3** – Has the fourth highest priority in case of congestion
- **Standard** - Has the lowest priority in case of congestion

A Committed Information Rate (CIR) and Peak Information Rate (PIR) are associated to each class. The bandwidth split between the classes is fixed. The CIR allocation per class is shown in the table below

Class	Classification Type			Bandwidth Allocation
	PCP, 802.1p	DSCP byte (IPv4 or IPv6)		
Premium	5, 6, 7	≥ 40	EF, TOS 5 , CS 5-7	15%
Business 1	4	32 - 39	AF41-43, CS 4	15%
Business 2	3	24 - 31	AF31-33, CS 3	20%
Business 3	2	16 - 23	AF21-23, CS 2	20%
Standard	0, 1	≤ 15	BE, AF11-13, CS 1	30%

Table 19. Class of Service details

The hierarchical nature of the CoS implementation means that bandwidth is shared across the classes as follows:

- The basic bandwidth allocation % (CIR) in the table reflects the minimum guaranteed per class
- Premium is strictly limited to 15% whereas Business 1 can use up to 30% (from unused Premium)
- B2, B3 and Standard can re-use any bandwidth that is unused by other classes so that B2, B3 and Standard can use up to 100%

If a customer wishes to use fewer than 5 classes then a subset of the available classes should be used as appropriate. The bandwidth sharing will allow the selected classes to fully utilise service bandwidth.

Example: A customer could opt to use 2 classes by selecting B1 (which can then use up to

30%) together with Standard (minimum 70%, maximum 100%)

The main purpose of the CoS service feature is to manage the bandwidth within the customer service itself. This is mainly relevant for Ethernet VPN where egress congestion may occur (“targeted overload”) and on the egress for overbooked Hubs.

The CoS mechanism would also be triggered if there was congestion at some point on the network . However this would be very rare because Colt operates an uncontended network.

Customers are responsible for setting up their traffic prioritisation rules and selecting the CoS classification required. The classification can be based on one of the following options:

- IP DSCP (IPv4 or IPv6)
- 802.1p marking

The following table summarises the availability Class of Service:

Access Type	Technology	CoS
Colt Fibre	Ethernet over MSP, Ethernet over NGN (IQNet)	Yes
ULL Fibre (Asia)	Ethernet over MSP, Ethernet over NGN (IQNet)	Yes
3rd Party Leased Line	Ethernet (with Colt Ethernet Demarcation Device)	Yes
3rd Party Leased Line	Ethernet (without Colt Ethernet Demarcation Device)	No
Unbundled local loop (ULL)	Ethernet First Mile (EFM)	Yes

Table 20. Ethernet Service CoS availability by access technology

More sophisticated CoS can also be provided using Colt CPE Solutions over a wider range of access types.

Class of Service is available for E-VPN on all platforms but is only currently available on the M-MSP platform for Ethernet Line and Ethernet Hub & Spoke.

Class of Service is not available for Private Ethernet.

9.3 Automated Monitoring and Notification (AMN)

Automated monitoring & Notification is the end to end capability to monitor Colt Network and the high volume services delivered on the Colt network. This enables Colt to provide automated proactive monitoring & notification by email to interested parties. These can be internal CRMs, AEs or customer contacts. Customer can manage their notification subscriptions directly from Colt Online.

Colt Ethernet services include a proactive notification service, which means that Colt proactively monitors the service and notify customers of any service-affecting faults (Service Down) via e-mail. In case of any network related faults, resolution is progressed actively until fault is resolved. If the fault is related to last mile, customer is notified and requested to confirm if the issue is related to any local activity or maintenance or else they want Colt to progress actively for resolution. Non Service affecting or Service degraded faults at last mile are currently not in scope of AMN. This is a standard option on the majority of Colt's Ethernet services, details are shown in the table below:

Access Type	Technology	Proactive Notification
Colt Fibre	Ethernet over IQNet Ethernet over MSP Ethernet over Fibre	Yes
ULL Fibre (Asia)	Ethernet over IQNet Ethernet over MSP	Yes
3rd Party Leased Line	Ethernet (with Colt Ethernet Demarcation Device)	Yes
3rd Party Leased Line	Ethernet (without Colt Ethernet Demarcation Device)	No
Unbundled local loop (ULL)	Ethernet First Mile (EFM)	Yes

Table 21. Proactive Notification Option Availability

Also note the differences in the way notification emails are sent to the customers :

- In Europe/North-America : as standard feature, Proactive fault notification email will be sent to customer **within two (2) minutes** for any service outage greater than **eight (8) consecutive minutes**.
- In Asia : as standard feature, Proactive fault notification email will be sent to customer **within 10 Minutes** for any service outage greater than **four (4) consecutive seconds.**

Any other requirement will require analysis via bespoke process.

9.4 Link Aggregation (LAG)

Colt's Ethernet services support link aggregation (LAG) as a standard customer option on 1Gbps and 10 Gbps interfaces. LAG is based on the Link Aggregation standard in IEEE 802.1AX-2008 (formerly defined in IEEE 802.3ad). This provides interface redundancy using two 1Gbps or 10 Gbps ports. LAG operates in active-standby mode only (IEEE 802.1AX-2008 Clause 5.6.1), traffic is not distributed over both ports. If one interface should fail, LAG automatically directs all traffic to the working interface.

LAG is not supported on 100G interfaces for Private Ethernet.

9.5 Demarcation Device (DD)

Demarcation Device is the device (L2 NTE) that is installed in customer premises when we deliver an offnet Ethernet service. It's the same equipment used to deliver on-net Ethernet services ie Accedian devices.

Depending on the customer segment, this feature is either delivered by default (Enterprise customers) or optional (Strategic Alliance customers). Note that it's not possible to opt out this feature for Enterprise customers. It's also mandatory to install a DD for Ethernet VPN offnet sites (see paragraph Off-Net E-VPN Access sites).

Demarcation Devices are required to be able to provide several features when an Ethernet service is delivered in an off-net site such as Automated Monitoring and Notification (AMN), Link Loss Forwarding (LLF), Performance Reporting, LAG and Class of Service (CoS). It's also useful for Service Assurance teams to help them troubleshoot service failures.

From a Service Delivery perspective, it's not the same process to deliver a DD in Colt countries or in non-Colt countries. It's always **bespoke to install a DD in non-Colt countries** (also called Global Reach countries) as it's difficult to have a global partner able to deliver these devices anywhere in the world: it's always *strongly* recommended to use an OLO who's able to deliver both the circuit and the DD at the same time.

Important note about DD delivery in non-Colt Countries:
Since Accedian has been bought out by Cisco, all the models have been rebranded by Cisco and they now have different part numbers/BOM even though they are pretty much the same equipment. However, as we did not onboard these new devices in our systems (Netcracker, SDN stack, monitoring tool, …), we're not able to use them and then we cannot use those provided by 3rd party providers in non-Colt countries. Therefore, for the time being, it's impossible to purchase and install a Demarcation Device in a customer offnet site. A bespoke solution must be raised for a workaround solution.
This comment does not apply for DD installation in GCR countries (for example in Poland or Romania in Eastern Europe) or in Colt countries.

Important note about Dubai:

Considering the current situation for DD installation described in the note above, Ethernet services are delivered by default without Demarcation Device in offnet sites in Dubai.

However, for scenarios where a Demarcation Device is required, it's still possible to use an existing workaround. In this case, Sales will have to follow the Container journey in CPQ and use the Excel pricing tool and an Ethernet EOF to place the order.
SEs will have to raise a bespoke solution but they can use an existing pre-approved bespoke document using the following reference ***SR_2873*** to implement the solution.
In this case, as no DD will be installed in the customer premise, it won't be possible to

provide proactive monitoring/ticketing. In the event of a service failure then the customer will have to raise a trouble ticket. This needs to be clearly advised to the customer during the presales phase. This will be also detailed in the pre-approved bespoke document.

If no DD is required, you can get directly a quote from Du or Etisalat that are the 2 local operators allowed to provide data services in Dubai and use the standard sales journey.

10 Resilience & Diversity

10.1 Architecture

The majority of Colt's platforms are built to a redundant and diverse design, thus ensuring that a wide range of resilience options can be provided to customers.

10.2 Resilience Options

Depending on the location, area and circuit speed, Colt Ethernet services can be provided as unprotected, backbone protected or fully protected.

Resilience options that are available for Ethernet Line, Ethernet Hub & Spoke and E-VPN include:

- **Unprotected –** Ethernet Line, Ethernet Hub & Spoke and E-VPN services are available with unprotected access where a single fibre extends from the Colt node to the customer premises. In this case, the Colt core network remains protected with primary and failover paths. This is the default configuration to connect retail buildings with the IQNet Platform (Star architecture).

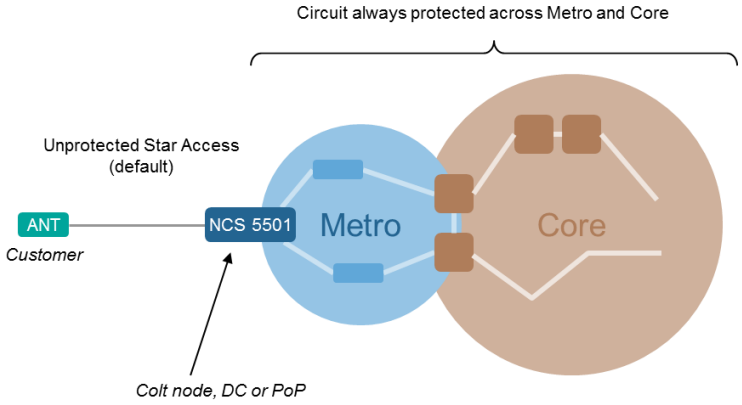


Figure 19. Unprotected Access on IQNet

- **Protected** – the service is protected by a fibre protection path in addition to the worker path, with automatic failover between fibre paths provided by protection switching as well as automatic restoration once the worker path is available

again.(This was the former default architecture to connect retail buidlings on MMSP platform which was based on fibre rings.)

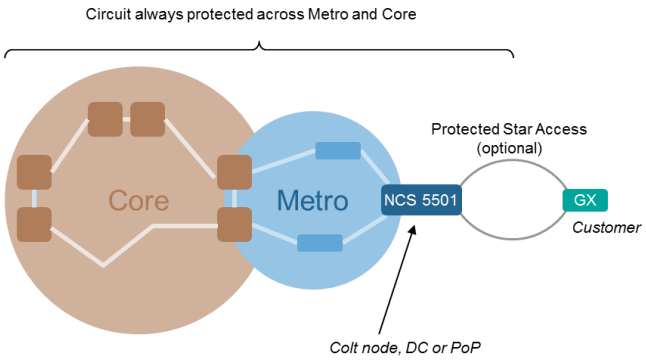


Figure 20. Protected Access on IQNet

- **Gold (Not available anymore – retained for informational purpose only)**

  Enables interface resilience by introducing Dual NTE (Colt terminating equipment) onto the same customer ring at both the A and B end of the service (former MMSP architecture) or by using a second fibre to deliver the second access (IQNet architecture). Both sub-services are protected in the IQNet metro and core but are not routed on diverse path (they can share the same device in the core/metro backbone).This adds full 1+1 NTE resilience into the solution architecture through duplication of the service with an additional customer interface. Failover between the services is a customer responsibility unless Colt CPE Solutions Enhanced Resilience is also deployed.

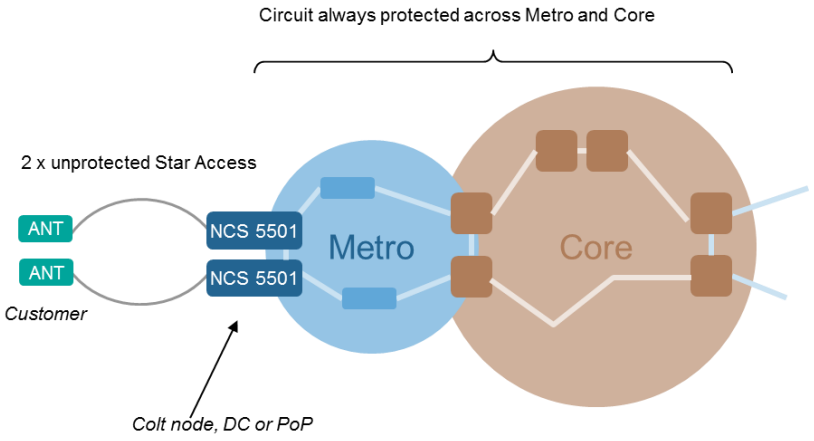


Figure 21. Gold Resilience Equivalence (Dual Access) on IQNet

- **Platinum (Not available anymore – retained for informational purpose only)**
  Similar to the Gold level but ensures the maximum level of protection that can be provided over Colt's infrastructure. By placing each network terminating equipment

on a different fibre ring (former MMSP architecture) or providing two protected access (IQnet architecture), this ensures a maximum of 1+1 end-to-end redundancy across the solution. Note that the two protected star access tails may share the same fibre path : it's only required that the two fibre path for each of the protected acces are diverse.

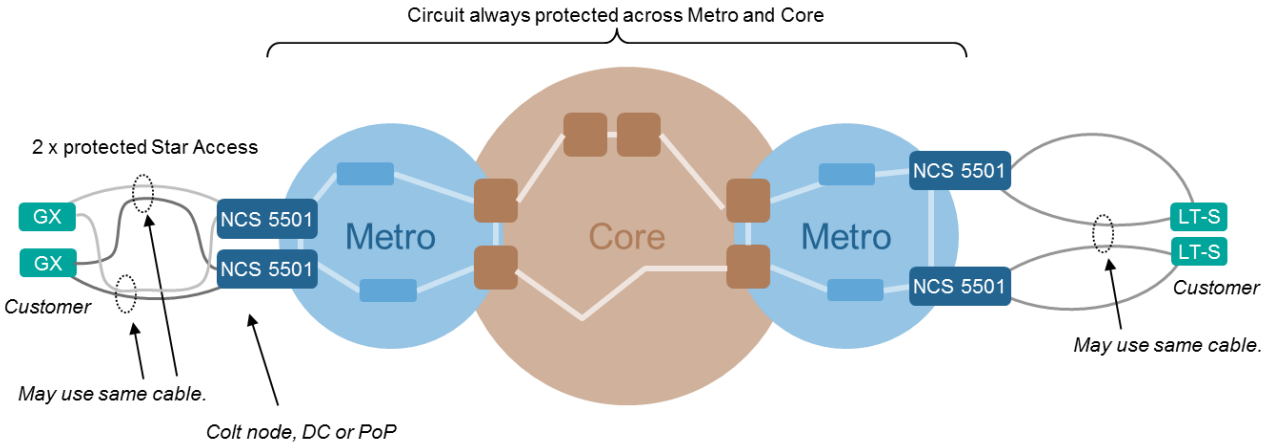


Figure 22. Platinum Resilience Equivalence (Dual Protected Access) on IQNet

An Ethernet Line or Ethernet Hub and Spoke solution can comprise sites with any combination of Gold and Platinum resilience options. Sites with Unprotected or Protected resilience cannot be combined with either Gold or Platinum resilience.

For Ethernet Hub and Spoke, it is possible, on request, to accommodate Standard resilience spokes with a Gold and Platinum hub. However, it should be noted that the standard spoke site is only then connected to one of the hub sites.

Private Ethernet services can be delivered as protected or unprotected :

- **Unprotected** – Private Ethernet is available as an unprotected metro service based on a single dedicated fibre at 1, 2, 3, 4, 5, 6Gbps and 10Gbps between the customer dedicated terminating CPE at each site.
- **Protected** –Private Ethernet service is available as a protected service based on 2 separate fibres extended between the customer dedicated terminating CPE at each site. One fibre acts as the worker path and the other fibre as the protection path. In the event of a fibre failure then the terminating CPE automatically switches traffic to the protection path with failover time <50ms



Figure 23. Private Ethernet resilience options

It should be noted that the Protected option do not mandate the absence of single points of failure neither on the terminating equipment nor in the Access/Backbone part in the standard service.

Customers are responsible for ensuring they have appropriate levels of resilience in their network to avoid serious service disruption in the event of an outage.

10.3 Diversity feature

10.3.1 Old version based on Types (no longer supported - informational only)

Additional resilience features that can be provided as part of an Ethernet service, in common with other parts of the Colt product portfolio include:

- **Diversity** – two services can be provided in a diverse configuration from one another, with each service following a designated network route if required. The diversity feature is classified into 4 sub-types which define different levels of diversity between two services and across all the network components that make up a service end-to-end. The type of diversity (1 to 4) depends on the Colt's assets available in the location where the customer asks for diversity.

Attributes / Products		*Diversity Type*			
Ethernet Resilience Level		Diversity 1 "Max"	Diversity 2 "Strong"	Diversity 3 "Partial"	Diversity 4 "Alternate"
A	NTE (=CPE)	Diverse	Diverse	Diverse	Diverse
B	Cabinet / Rack	Diverse*	Diverse*	Diverse*	Diverse*
C	Customer Power Source	Diverse*	Diverse*	Diverse*	Diverse*
D	Colt Power Equipment	Diverse	Diverse	Diverse	Diverse
E	Internal Cabling/ Long Line	Diverse*	Diverse*	Diverse*	Diverse*
F	Dual Building Entry	Supported*	Supported*	Supported*	Supported*
G	Physical Path (Metro)	Diverse #	Diverse #	Diverse #	Diverse #
H	Fibre (Metro) (The fibre itself, not the physical path)	Diverse	Diverse*	Diverse*	Diverse
I	Homing Node / PE Equipment (N/A for EoFibre Metro)	Diverse	Redundant	Single	Diverse
J	Node Diversity (N/A for EoFibre Metro)	Diverse	No	No	Diverse

Table 22. Table of the Diversity types

* Building dependant.

10.3.2 New Diversity feature

Two services can be provided in a diverse configuration from one another, with each service following a designated network route if required. The standard Diversity feature defines different levels of diversity between two services and across the access components. The standard Diversity depends on the Colt's assets available in the location where the customer asks for diversity.

The standard Diversity is available on both IQNet and M-MSP platform even if the IQNet platform is the preferred one. The list of the cities where the standard Diversity applies can be found here.

Please have a look at the Ethernet Diversity Guide on the Intranet too.

Definition of Standard Diversity for Retail Buildings (Colt Fibre) :

- **2 x NTE in the customer premise** : by default, the NTEs will be delivered in the same rack unless the customer requires to install them in different racks and they will have only one power feed. The power source will be the same unless the customer is able to provide two different sources. When Internal Cabling or Long lining is required, by default, it won't be diverse unless it's requested by the customer.
- **2 diverse fibre paths between the RB and the Colt POP** (Colt node or third party DC). When Diversity is ordered for Protected circuits, by default, two diverse fibre path must be provided for the worker path of each circuit or the worker and backup path. On the IQNet platform, the nodes used to connect the RBs are called "Star Aggregation Sites".
- **2 diverse homing node (HN)/access router (AR)** either located in the same Node or in two different Nodes. By default, the HN/AR will be located in the same rack if they're in the same building. The HN refers to the device used in the M-MSP environment while AR refers to the Cisco 55xx series used in the IQNet platform.

  ***Note** : In some cities, just one Star Aggregation Site with one AR is available and then the installation of a second AR (Cisco 5501) will be triggered with the order but will generate a 3 month lead time. It's possible to raise a Capacity check to know if a 2nd AR was previously installed and if some capacity remains avoiding the 3 month delivery lead time.*

Definition of Standard Diversity for Retail Buildings (OLO tail) :

- **2 x NTE in the customer premise** : the OLO must install a NTE for each circuit delivered even though the same OLO is used to provide the two circuits. When Internal Cabling or Long lining is required, by default, it won't be diverse unless it's requested by the customer. Note that Colt Demarcation device can be installed on top of the OLO NTE.

- **2 circuits between the off-net RB and the delivery POP** (POP where the OLO delivers the service to Colt ie the external NNI). ***Please note that in order to meet the Diversity definition, it's not mandatory to deliver diverse circuits as Colt does not own the fibres.*** However, some operators like Orange or BT have options in their portfolio to deliver diverse circuits and should be used whenever requested by the customer. It's also possible to use 2 different OLOs but in this case, the diversity of the fibre paths is almost impossible to deliver as the providers may use the same ducts or buy from another operator to deliver the service hence overlapping circuits may happen and creates SPOF.
- **2 diverse circuit handoff devices** either located in the same delivery POP or in two different ones. By default, these devices can be located the same rack if they're in the same POP.

  It's also possible to provide the Diversity option in DCs. In this case, it would mean that there is no real access section for example or the services may not be delivered on NTEs but on a patch panel in a MMR.

Definition of Standard Diversity for Data Centres :

- **2 x customer hand-off :** by default, the customers hand off will be delivered in the same MMR/customer rack if applicable unless the customer requires to deliver his service in diverse MMR/customer racks in the same DC. If the service is delivered by using NTE, the rules for NTE when we deliver in a RB apply.
- **2 diverse homing node (HN)/access router (AR)** either located in the same Node or in two different Nodes. By default, the homing nodes will be located in the same rack if they're in the same building.

  ***Note** : there may not be an access part when the service is delivered in a DC but most of the DCs having just one AR, one of the circuits will have to be extended in another Colt POP/Node using the Colt fibre (Eg. London) to provide the diversity of this device. In some cities, just one node with one AR is available and then the installation of a second AR (Cisco 5501) will be triggered with the order but will generate a 3 month lead time. It's possible to raise a Capacity check to know if a 2nd AR was previously installed and if some capacity remains avoiding the 3 month delivery lead time.*

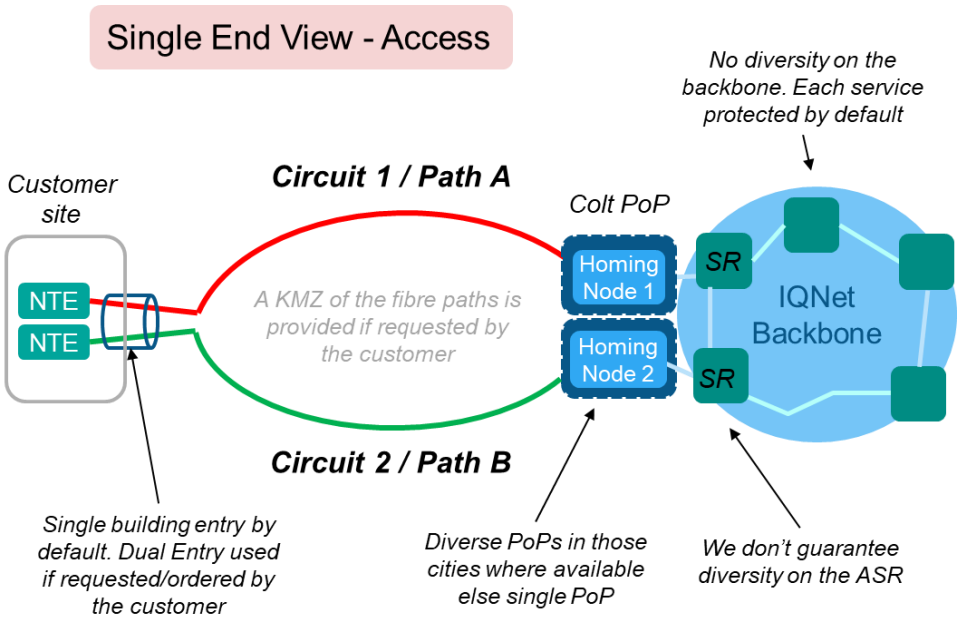


Figure 24 : Standard Diversity delivered on IQNet for a RB (on-net)

The Diversity always applies on two circuits and therefore when ordered at the same time, each one will have its own order form with the Diversity option selected on one of them.

The Diversity is ordered by choosing the option (Yes/No) on the order form whether it is embedded in CPQ/eOrderLite or is a separate EOF (CPQ container). The reference of the other circuit it needs to be diverse from must be specified as it is today.

In the case the Standard Diversity option fits to the customer then it can follow the Transactionnal process without engaging a Sales Engineer.

Otherwise, if it does not meet the customer requirements because he wants something that exceeds what we can deliver then a Sales Engineer must be engaged : he will decides if the solution can be designed with standard components or if he needs to raise a Solution Request. **In both cases, Service Delivery cannot reject the Diversity option when selected even if it does not meet the Definition as long as it's documented either with a Solution Design Document or a SR.**

From a pricing perspective, +20% will be charged on top of the base NRC/MRC of the service when the Diversity option is ordered as a Standard.

The Diversity has also an impact on the SLA (availability). It is possible to have a combined SLA for both circuits (99.999% - see the SLA document) under certain circumstances or to keep a SLA per circuit.

Important notes to consider about Diversity :

Note 1 : *Diversity of fibre path is subject to fibre availability and therefore can only be confirmed at the delivery stage. We expect that this will be fine most of the times as we have a large fibre infrastructure but such cases can happen in the extremities of our MANs. In this case, SD will raise the issue to the Order Manager/Sales in charge of the customer to find a*

solution to provide the fibre path diversity. This might involve additional unforeseen costs.

Note 2 : *if a customer requests a KMZ of the access part of the circuit, the request must be raised by the Sales person at the early stage of the sales cycle to provide it in time. Bear in mind that this document is purely indicative as Fibre Planning teams don't book the fibre at the presales stage. Therefore there is no guarantee that, at the moment of the order, the circuits will be deployed as per the KMZ and this should be clearly stated in a caveat when we provide a KMZ to a customer. However, the KMZ file will have to be attached to the order for SD teams to try to deliver accordingly. KMZ are only available for circuits strictly greater than 100 Mbps.*

Note 3 *: When using OLO tails, It may not be always possible to get a KMZ from the OLO.*

Note 4: *on the IQNet platform, the nodes that are used to connect Retail buildings when using Colt Fibre are called "Star Aggregation Sites". The IQNet location finder tool (https://dcp.colt.net/dclocator/) can be used to locate them in every cities in Europe and to check if several Star Aggregation Sites are available in a given city. (This tool is also called Star Aggregation Site locator).*

Note 5 : *the Diversity for Ethernet services is only available on the access part of the circuit. The diversity cannot be provided on the backbone part including the SR (Aggregation Service Router – Cisco 9K) : there is only one SR in some cities. You can check SR's availability with the Star Aggregation Site locator*

Note 6 : *in some Key DCs, the Diversity feature only applies to the AR and the handoff (or the NTE when available) as there is no "real" access circuit. Standard DCs are managed like Retail buildings from an access point of view*

Note 7 : *the Single Entry will be used by default and in this case, the fibre path cannot be diverse close to the building entry. However if a customer requests to use a Dual Entry, it must be specified on the order form as a separate component of the whole Diversity solution. If a Dual Entry is already existing, Sales/SE will have to check if an Internal Cabling is required to use it and charge for it as per the BAU process, otherwise, a BCP reference must be used with the Dual Entry order.*

Details about the standard Diversity architecture:

1) Ethernet Line between Retail Buildings (IQNet) – Colt Fibre:

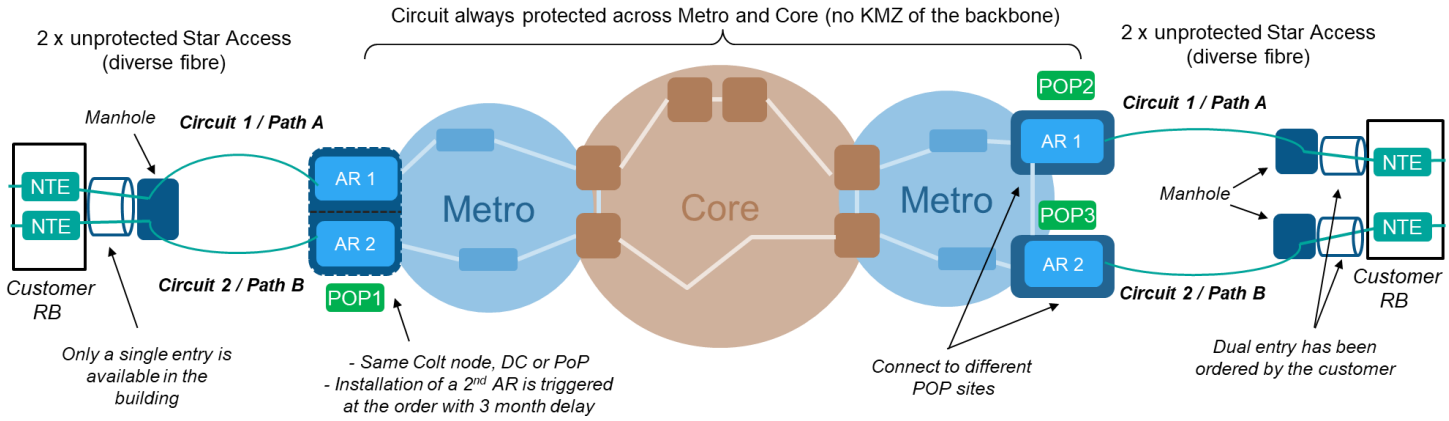


Figure 25 : Diversity architecture on the IQNet Platform

- The Diversity only applies on the Access part.
- IQNet is the preferred platform to deliver Diversity as stated in the Deployment rules.

2) Ethernet Line between Retail Buildings (M-MSP) – Colt Fibre:

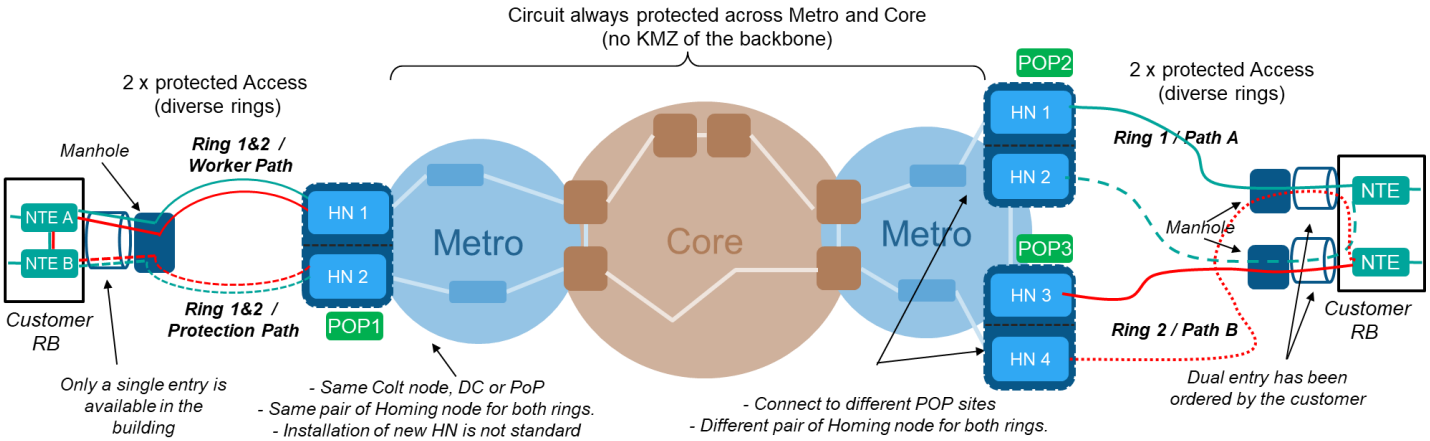


Figure 26 : Diversity architecture on the MMSP Platform

- The Diversity only applies on the Access part.
- Two rings are used for both services. MMSP should be used only if no other option or for specific reasons as this platform is reaching resource limits in some areas.
- The worker path of both rings will be on the same fibre path hence no diversity between them and it's the same for the protection path. The diversity is provided between Worker Path and Protection Path only.
- This means that the active path of both circuits may be impacted in case of failure on the fibre cable where both active path are running but they will switch over the traffic on the backup path automatically.
- Cities where HN diversity is not available, are considered as Bespoke as we don't

want to invest in the M-MSP platform to add another HN.

3) Ethernet Line between Retail Buildings (M-MSP/IQNet) – OLO tail:

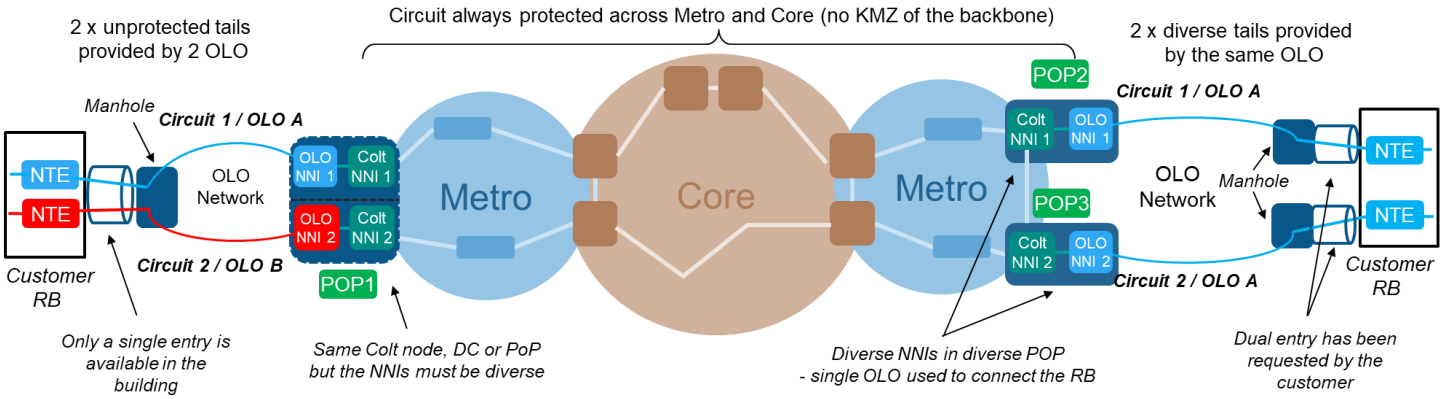


Figure 27 : Diversity architecture on the IQNet/MMSP Platform with OLO tails

- The Diversity only applies on the Access part.
- Default Diversity is available when using OLO tails and does not necessarily require SE engagement.
- When using OLO tails, it's not guranteed to deliver Diverse access circuits.

4) Ethernet Line between Standard DCs (IQNet/M-MSP):

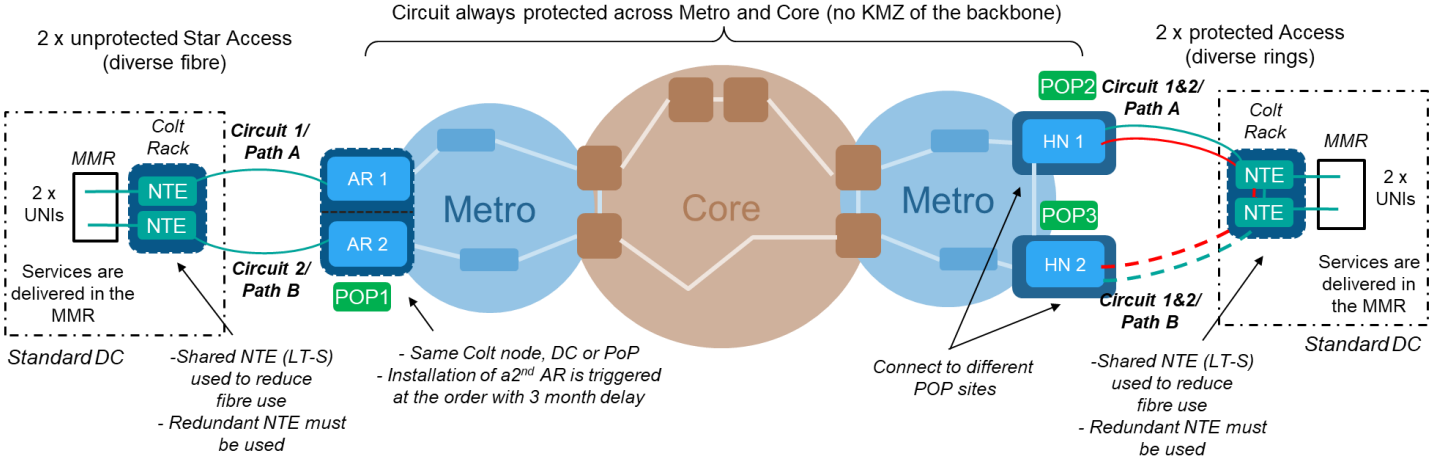


Figure 28 : Diversity architecture on the MMSP/IQNet Platform

- The Diversity only applies on the Access part.
- Standard DCs can be considered as Retail Buildings from an access point of view.
- Same comment applies to the MMSP rings in this case : Diversity will be provided only between the Worker paths and the Protection paths of both rings.

- The use of shared NTE in Colt's racks is up to the SD team decision as long as Standard Diversity criteria are met.

5) Ethernet Line between Key DCs (IQNet)

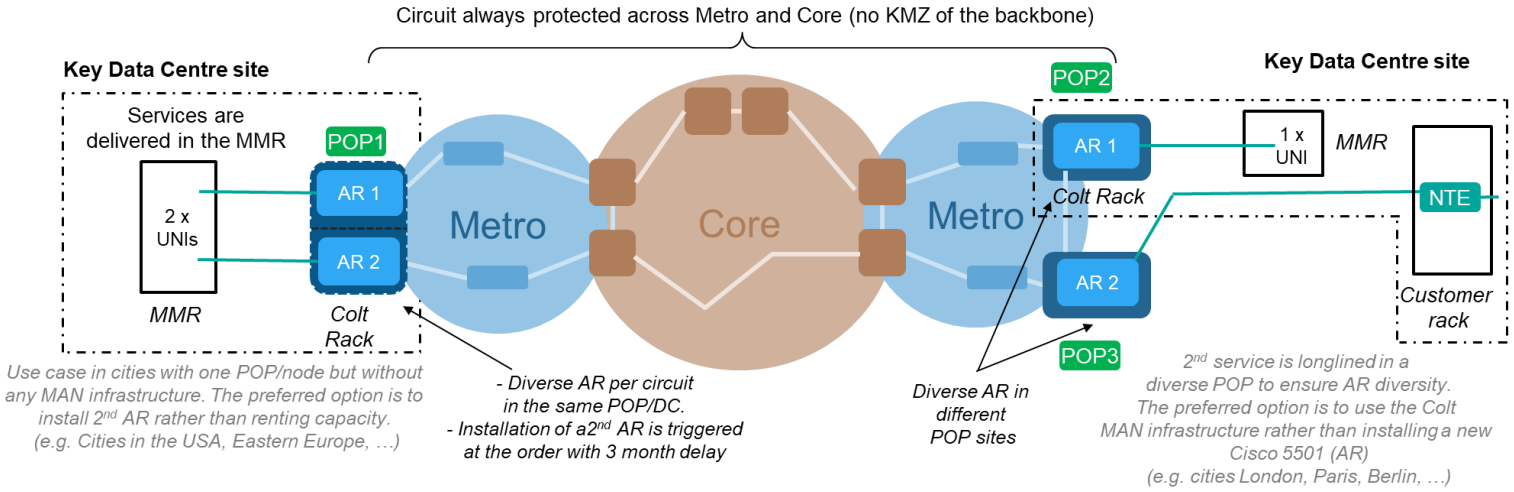


Figure 29 : Diversity architecture to connect Key DCs

- The Diversity only applies on the Access part.
- During the IQNet roll-out, only one AR (Cisco 55xx) has been installed in the POP because of budgetary constraints. Therefore in some scenarios, a second Cisco 5501 must be installed in the IQNet POP to provide AR diversity : the standard lead time is 3 month in this case.
- The concerned scenarios are those where we don't have a MAN ie no Star Aggregation Sites (USA, eastern Europe) or just one Star Aggregation Site (eg. Hanover). It's mentioned in the « Standard Diversity list » file.
- In the IQNet cities where we have a MAN, the preferred option is to extend the service from another IQNet POP to get the AR diversity as shown on the diagram above (right hand side of the diagram).

10.3.3 Other options coming along with the Diversity option :

- **Dual Customer Power Source – Not available as standard**
  Where a customer has dual/diverse power supplies available in proximity to one another, they may direct that two Colt services should be connected to different power sources. Providing redundant main power sources within environment specified for service termination by the customer is a customer responsibility. In this case, Colt is able to deliver devices with dual feeds on a bespoke basis. The devices that are delivered do not have on-board dual power sources and by default, a single power feed. See the "11.1 Network Termination Equipement specifications" section

for more details about the power feeds.

- **Dual Building Entry** – Establish a second building entry point for the first time or indicate that a service order should use existing multiple entry points.

11 Service Presentation

11.1 Network Termination Equipment (NTE) specifications

Colt uses a number of technology platforms to deliver Ethernet services. The platform used by an individual service will vary depending upon the signal type, bandwidth, access method and other factors. It will be in accordance with Colt's deployment rules for Ethernet services and be sufficient to provision the circuit to the correct specification.

Equipment	GT / GT-S	LT-S	GX
Appearance	 (This picture is GT-S)		
Services	Ethernet Line/Hub & Spoke/Ethernet VPN	Ethernet Line/Hub & Spoke/Ethernet VPN/Private Ethernet	Ethernet Line/Hub & Spoke/Ethernet VPN
Platform	MMSP	MMSP / IQNet	MMSP / IQNet
Used for	2Mbps - 1Gbps	2Mbps - 10Gbps	2Mbps - 1Gbps
Size	38mm(H) x 200mm(W) x 172mm(D)	45mm(H) x 330mm(W) x 225mm(D)	35mm(H) x 200mm(W) x 172mm(D)
Rack Unit	1RU (Rack mount)	1RU (Rack mount)	1RU (Rack mount)
Power	100～240 VAC / 20～57 VDC	100～240 VAC / 20～57 VDC	100～240 VAC / 20～57 VDC
Max Power Consumption	27W	90W	40W
Weight	1.35kg	2.7kg	1.35kg
Interface	GT: 10/100/1000BASE-T GT-S : 10/100/1000BASE-T 1000BASE-SX/LX	10/100/1000BASE-T 1000BASE-SX/LX 10GBASE-SR/LR	100/1000BASE-T 1000BASE-SX/LX
Max Number of Interfaces	GT: RJ-45 MAX 2 Ports GT-S: MAX 2 Ports (RJ-45 or LC Connector)	LC Connector (10G) 2 Ports LC Connector（1G） Max 8 Ports RJ-45 Max 4 Port (Exclusive use with LC)	RJ-45 Max 4 Port LC Connector（1G） Max 4 Ports

Operating Environment	Temp：0 - 50℃	Temp：0 - 50℃	Temp：0 - 50℃

Equipment	**LX**	**ANT (1G)**	**ANT (10G)**
Appearance			
Services	Ethernet Line/Hub & Spoke/Ethernet VPN	Ethernet Line/Hub & Spoke/Ethernet VPN	Ethernet Line/Hub & Spoke/Ethernet VPN
Platform	MMSP / IQNet	IQNet	IQNet
Used for	2Mbps – 10Gbps	2Mbps - 1Gbps	2Mbps – 10Gbps
Size	38mm(H) x 229mm(W) x 200mm(D)	36.1mm(H) x 118.6mm(W) x 133.4mm(D)	38.1mm(H) x 142.2mm(W) x 203.2mm(D)
Rack Unit	1RU (Rack mount)	1RU (Rack mount)	1RU (Rack mount)
Power	100 - 240 VAC / 20 - 57 VDC	100 - 240 VAC	100-240 VAC
Max Power Consumption	55W	40W	20W
Weight	1.66kg	625g	100g
Interface	1000BASE-SX/LX 10GBASE-SR/LR	1000BASE-SX/LX/T	10GBase SR/LR
Max Number of Interface	Max: 4 ports (SFP+ Optical)	Max 1 customer port (RJ-45 or LC Connector)	Max 1 customer port (SFP+ Optical, LC)
Operating Environment	Temp:0 -50℃	Temp：0−50℃ or -40～65℃ (with fan)	Temp：0-50℃ or -40～65℃ (with fan)

Equipment	**NCS540**	**NCS 5501 / Cube C-2508***	
Appearance			
Services	Private Ethernet and Ethernet Line	Ethernet Line/Hub & Spoke/Ethernet VPN	
Platform	IQNet	IQNet	
Used for	20G – 100G	20Gbps - 40Gbps	
Size	43.7mm(H) x 440mm(W) x 250mm(D)	43.7mm(H) x 442mm(W) x 551.2mm(D)	43.5mm(H) x 483mm(W) x 230mm(D)
Rack Unit	1RU (Rack mount)	1RU (Rack mount)	1RU (Rack mount)
Power	90～265 VAC / -40～-72VDC	90～265 VAC / -40～-72VDC	100～240 VAC / -48VDC
Max Power Consumption	250W	330W	40W
Weight	4.7kg	10.35kg	4.5kg
Interface	100GBASE-LR4	100GBASE-SR10/LR4	-

Max Number of Interface	Max 2 x 100G customer ports (LC Connector)	MPO/MTP-24 for SR10: max 1 port LC or SC connector for LR4: max 1 port	-
Operating Environment	Temp： -5～55℃	Temp : 0～40℃	Temp : 0～50℃

* Cube is used when we put a ULL fibre at the customer site in Asia until the 40km transceiver is supported.

Table 23. Network Termination Equipment

Note about NTE power feeds and power supply:

All the NTE that we install in customers' premises have only one in-built power supply but support single/dual power feed. This allows two external power sources to be used for resilience purposes.

By default, we install single power feed devices however it is possible to deliver NTEs with dual power feed if specifically requested by the customer. In some cases where only an AC power source is available it may be necessary to deploy external rectifiers for AC/DC conversion in order to support specific models.

For example, for the Accedian GX device, we use both the AC single power feed (default) and the DC dual power feed models. When single AC input is required we use the AC model while for any other power feed configuration (single DC, dual DC or dual AC) we use the DC dual power feed model. Therefore in a configuration where power feed redundancy is required and dual AC power is available, the set up would be:

- 1 x Accedian GX – DC dual power feed model
- 1 x Y-cable
- 2 x external AC/DC rectifier

Note : If NTE devices other than the default are required for a specific customer (e.g. dual power feed device is requested) then this must be recorded in the *Notes* that accompany the order.

11.2 Auto-Negotiation

Auto-Negotiation (Auto-Neg) is a standard handshake mechanism for the negotiation of attributes between interfaces. Attributes include elements such as interface rate and duplex mechanism. When configuring services **Colt sets Auto-Neg = ON by default** for electrical and optical interfaces. In cases where customers require Auto-Neg to be switched off then this is possible for some interfaces as indicated in the table below.

Device	Interface			
	100BaseT	1000BaseT	1000BaseSX/LX	10GBaseSR/LR
Accedian (ANT, GX, LT-S, LX)	ON/OFF	ON*	ON/OFF	N/A
Cisco NCS	ON/OFF	ON*	ON/OFF	N/A

Table 24. Auto-Neg Interface Settings

* Auto-Neg setting for 1000BaseT (1GbE) electrical interface is set = ON and is mandatory.

If a customer requires Auto-Neg to be disabled then this needs to be requested and communicated at time of order entry.

11.3 Demarcation point

The service demarcation point for Colt services is the customer interface of the Colt access equipment. Colt will normally present its Ethernet services on an industry compliant patch panel within a Colt cabinet but may also present this within a customer cabinet or designated building 'meet me room' common equipment area. The precise customer site deployment will normally be determined by a site survey during the service delivery phase and will be influenced by the technical deployment rules of Colt or its partners, the cabling architecture within the customer building and the customer's equipment area. The deployment may typically include an Ethernet NTE, patch panel and power conditioning equipment.

The demarcation point will be labelled so that the customer can identify each unique service on the patch panel. This label will correspond to that within the Colt Service Handover Document.

If in-house cabling is required from the Colt cabinet to customer equipment, this will be a chargeable service. If Colt installs cabling on the customer's behalf, the demarcation point remains the same: the base of the Colt cabinet. Colt is not responsible for fault-finding on the in-house cabling.

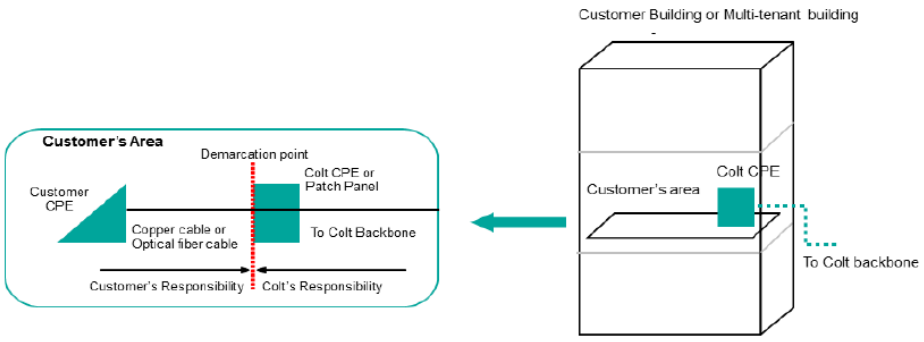


Figure 30. Demarcation Point

11.4 Power

Precise power requirements will be identified and planned during the service delivery phase.

The customer must provide one or more AC mains power outlets in accordance with the power regulations pertaining in the relevant jurisdiction. It is the customer's responsibility to ensure that the power supply meets electrical standards, is fused and safe to use by Colt.

Under circumstances where Colt locates its equipment in a customer cabinet, Colt may use DC power if this is already provided. This must conform to -48V DC and be able to provide suitable connectors as required by Colt equipment.

11.5 Environmental Conditions

The customer environment must conform to the following environmental conditions:

Temperature: +5ºC - +40ºC

Humidity: 5% - 85%

12 Security

12.1 Colt

The physical security of our buildings is tightly controlled and access is strictly limited to authorised personnel only. All areas within Colt buildings are secured by means of an electronic access control system to ensure that access is controlled. All people must hold an appropriate pass card while on Colt premises. Non-Colt personnel are not allowed on Colt premises without specific authorisation and prior arrangement. Guests must be vouched for by a Colt host or verified by security in buildings.

12.2 Ethernet Services

Because Colt Ethernet services are separated at the OSI physical transmission layer, it is impossible for customers to intercept or send data to or from any other customer sites via their own access. Colt's Ethernet services are delivered utilising a network resource that is reserved for that service and for each individual customer.

13 Service Delivery

13.1 Service Delivery Timeframes in all locations for retail buildings

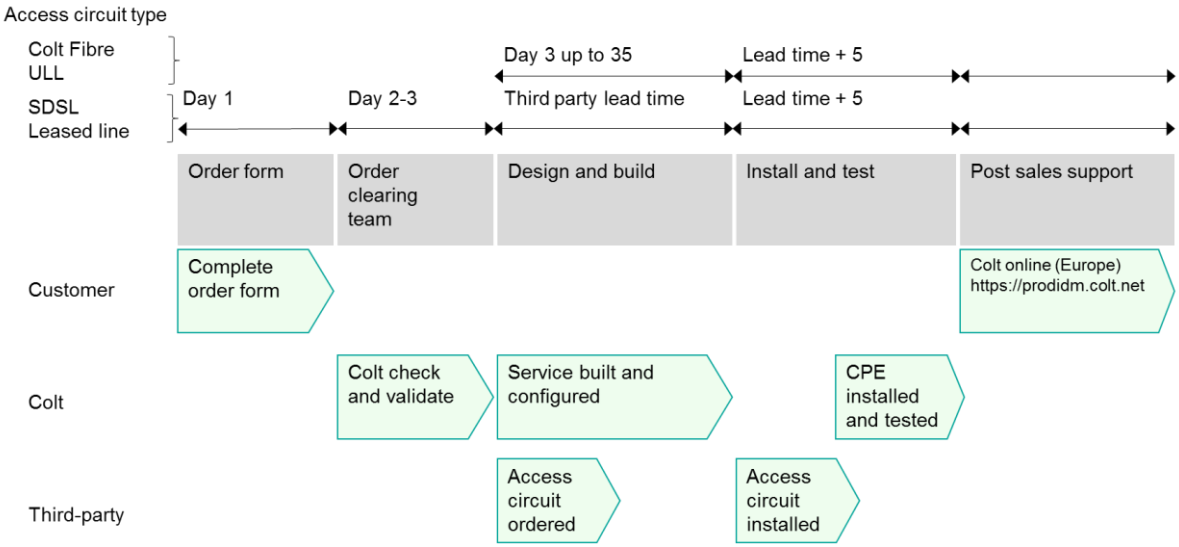


Figure 31. SD timeframes in retail buildings (working days)

Colt provides different target delivery lead times depending on the service to deliver. These values are indicative and the diagram above illustrates some of them. The SLA document includes a full description of the target lead times.

13.2 Service Delivery Timeframes in all locations for Data centres

Some of our Data centres are precabled and a specific process is in place to always ensure enough capacity to deliver more quickly our services : these are our Key Data centres on which we can target faster lead times than in standard data centres.

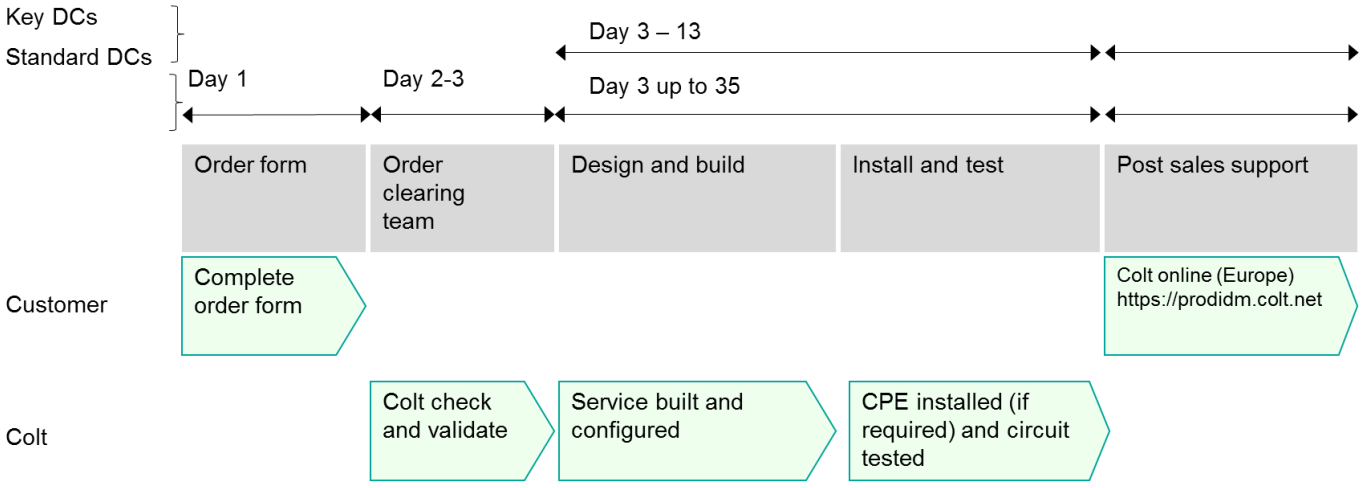


Figure 32. SD timeframes in Data centres (working days)

Colt provides several target delivery lead times depending on the service to deliver. These values are indicative and the diagram above illustrates some of them. The SLA document includes a full description of the target lead times.

13.3 New service order

A new service order is the initial provision of the service to the customer premises, including the initial configuration of the network to the specification provided in the order form.

13.4 Modifying an existing service

Modifying an existing service consists of the subsequent enabling or disabling of service features, functions and interfaces as well as service changes following initial installation, which are chargeable items.

13.5 Modifying an Ethernet service

Following provision and installation of an Ethernet service, customers can ask Colt to enable additional features, functions and interfaces, and request changes to the service. The implementation of changes is chargeable, and some changes may mean that revised rental charges apply. Changes fall into two categories: A and B.

Category A modifications - Regarded as a new provision in terms of lead times. These are change requests that require physical alternations to the equipment that delivers the service. Examples include changing the interface presentation or adding protection, which is regarded as a new provision in terms of lead times.

Category B modifications - Require 10 working days lead time. These are configuration changes that can be carried out remotely and within normal working hours and include the following:

- Changing the network path where capacity is already available
- Modifying the service presentation between options that are of the same bandwidth and are supported by the same interface card and cabling types

13.5.1 Out-of-hours changes

Changes that need to be handled out-of-hours must be scheduled and approved in advance, and there is a two-week lead time for scheduling them. (This does not apply to emergency changes). This is a chargeable option except in Japan where it’s free of charge.

13.5.2 Cessation of service

Cessation of service is within 60 working days from request by the customer. Request for cessation of service may be subject to a charge in accordance with Colt standard terms and conditions. Should the customer cancel their order during installation, Colt reserves the right to raise a charge.

13.6 Fast track installation

Fast Track allows customers to expedite delivery of their order and receive their service on a date specified by the customer (generally prior to the standard lead time), or, if this date is not possible, on the earliest possible date.

Fast Track installation is available for all On-Net sites in all Colt geographic locations except North-America, Japan and Singapore. If Fast Track is possible, Colt will provide a Fast Track delivery date based on technical feasibility, which customers can accept or reject. If accepted, the Fast Track option incurs an extra charge in addition to the standard installation fee.

13.7 Service testing

As IQNET operates on a shared network, the following service testing process must be followed:

- All services are tested remotely via the STP portal by running Y.1564 tests that are more advanced than RFC2544.
- The STP portal already has logic in place to ensure that testing is performed with a maximum configured bandwidth of 10G. This is the standard testing approach, and this is what is provided to customers by default.
- If the service bandwidth exceeds 10G, it is not supported via the STP portal. However, if the customer requests testing above 10 Gbps, this is only possible using a physical tester and sending technicians onsite. The test must be conducted during the customer's maintenance window and outside business hours to avoid any network congestion caused by the test traffic.
- Anything that falls outside of the standard process described above will be chargeable.

Note: This is not applicable for ELAN services (Ethernet VPN). For E-LAN tests, STP portal is configured only with 1Mb (EIR).

14 Service assurance

Colt provides a high level of service assurance:

- The core network is proactively monitored
- A local language help desk is available 24 x 7
- Colt Online provides a web-based portal that enables customers to view bills and trouble tickets

14.1 Customer service

Colt has a high quality fibre network that enables the provision of an annual target service availability. The target availability depends on the service taken and the location of customer sites. The fault help desk is available 24 hours a day, seven days a week. Customers can report a fault at any time by contacting the Customer Service Centre and speaking to a representative in their local language (Japanese and English in Asia).

When service is provisioned, customers are issued with a unique service reference for each circuit that they should always use when reporting faults. The contact number for fault reporting is specified in the handover pack and can also be found on the Colt support page for each country : http://www.colt.net/support/

14.2 Colt Online

Colt Online is an intuitive, user-friendly application enabling new and existing Colt customers to interact with Colt via a secure Internet connection without the need to speak to a Customer Service Agent or Account Executive.

Every Colt Online customer is provided with an administrator account for a defined user within their organisation. This administrator has full access to the available features for all their customer accounts and sub-accounts, including:

- Search and view any bill from the previous six months in .pdf format (not available in Switzerland due to data protection legislation)
- View the status of any order in the delivery process
- View the status of any ticket (covering faults, enquiries, service requests) in real-time
- Search and view all live services
- View an account dashboard, summarising the four features above

14.3 Services management

Colt Ethernet Services are offered as managed services. Management information is extracted from Colt's on-site equipment using a network management link to a Colt Network Management Centre (NMC). This enables Colt to carry out remote monitoring, configuration, troubleshooting and testing and reduces the need to send technicians to the customer site.

Colt has a high quality fibre network that enables the provision of an annual target service availability. The target availability depends on the service taken and the location of customer sites. The fault help desk is available 24 hours a day, seven days a week. Customers can report a fault at any time by contacting the Customer Service Centre and speaking to a representative in their local language (Japanese and English in Asia).

When service is provisioned, customers are issued with a unique service reference for each circuit that they should always use when reporting faults. The contact number for fault reporting is specified in the handover pack and can also be found on the Colt support page for each country : http://www.colt.net/support/

15 Commercials

15.1 Billing

The customer bill is available on paper or CD-ROM and details:

- Service installation charges
- Service rental charges
- Other charges and credits
- Discounts, where applicable

Customers are provided with a single freephone point of contact for all service enquiries including billing enquiries. For detailed billing enquiries, customers can be transferred to a specialised billing department.

Charges will be applied per site based on the bandwidth selected and the location of the site.

15.2 Installation charges

One-off installation charges will be billed in full in the first invoice issued after installation. This will usually be within 30 days of service delivery. In addition, any moves, changes or modifications will incur one-off charges.

15.3 Rental charges

Customers can opt to pay monthly, quarterly or annually. Where no option is indicated, monthly billing will be set as default. All rental owed under the service will be payable in advance.

15.4 Additional features

The charges for enhanced resilience options are built into the service charges.

16 Performance Objectives

Colt commits on a set of elements that represents the quality of the services delivered (lead times, TTR, performance, etc,…). Faults can be reported via the local telephone numbers, 24 hours a day, 7 days a week. The period when the service was not available will be measured from the time a trouble ticket is opened to the point Colt resolves the problem.

These are defined in full in the SLA document, which is a separate document and includes details of the rebates if the targets are not met.

17 Certifications and industry standards

Colt is dedicated to ensuring that our management systems adhere to the widely accepted International Standards Organisation (ISO) and British Standards Institute (BSI) standards. Colt holds the following certifications:

- **ISO 14001** - Internationally accepted standard that sets out a framework of essential elements for putting an effective Environmental Management System (EMS) in place. The standard is designed to address the delicate balance between maintaining profitability and reducing environmental impact. This certificate is held for all Colt countries and helps us to identify the impacts that our operations have on the environment and then plan how we will reduce our most significant ones. It also ensures that we comply with all environmental regulations in each country we operate
- **ISO 9001** - The world's most established quality framework that sets the standard not only for quality management systems, but management systems in general. Colt holds this certification for the provision of service management for all Colt Data, Voice and Network Services, plus management of the Colt core network, backbone, switches, routers, infrastructure and associated systems
- **ISO/IEC 27001** - The only auditable international standard which defines the requirements for an Information Security Management System (ISMS). The standard is designed to ensure the selection of adequate and proportionate security controls have been established and also formally specifies a management system Information Security Management System (ISMS) that is intended to bring information security under explicit management control. Colt holds this certification for our Customer Managed Service (CMS) Solutions from European Data Centres. This includes customer European network monitoring, management and support services. In Colt India, this includes billing, revenue services and the Financial Shared Service Centre (FSSC). In Colt Spain, this includes Colocation Services in non-Data Centre locations.